Highlights

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An all-Mach IMEX scheme for the six-equation two-phase model with unified mechanical, thermal and chemical relaxation at arbitrary rates*

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ABSTRACT

We present a second-order semi-implicit finite volume scheme for the single-velocity six-equation two-phase model with total energy, for stiffened-gas and Noble–Abel stiffened-gas phases with mechanical, thermal and chemical relaxation and for Mie–Grüneisen phases with instantaneous mechanical relaxation. The acoustic waves at low Mach numbers and the relaxation up to infinite rates are treated in a single implicit step: the pressure work and the three relaxation processes, a slower process being active only when the faster ones are instantaneous, enter one local closure per cell, solved by scalar root finding, and each stage requires a Newton solve of a nonlinear system for the cell pressures, with a time step restricted by the flow velocity only. At fixed mesh, under non-degeneracy assumptions and for data on the equilibrium manifold, the scheme converges to the corresponding equilibrium schemes as the relaxation rates tend to infinity, and it is formally asymptotic preserving in the low Mach number limit. At shocks, the defect between the conservative and non-conservative mixture internal energies is shared between the phases according to the shock relations of the model, uniquely for weak shocks and by shooting for strong ones, without calibrated heat exchange. All quantities are transported explicitly by a path-conservative Rusanov scheme with the material speed, and the implicit acoustic dissipation vanishes at low Mach numbers. A stiffly accurate implicit–explicit (IMEX) Runge–Kutta scheme gives second order in time. We prove that the scheme preserves moving contacts between stiffened-gas or Noble–Abel stiffened-gas phases, at uniform temperature when thermal or chemical relaxation is active. Numerical results in one and two dimensions, from the incompressible limit to strong shocks, cavitation and phase transition, confirm the accuracy and robustness of the method.

1. Introduction

The single-velocity six-equation two-phase model with relaxation [67, 68, 78] is widely used for compressible liquid–gas and liquid–vapour flows: cavitation in hydraulic machinery and fuel injectors, flashing and evaporation fronts and the rapid depressurisation of pressurised water in nuclear reactor safety, underwater explosions, and the shock loading of condensed materials and mixtures [6, 21, 25, 26, 69, 76, 77, 79]. Two features make these flows difficult. First, the Mach number varies by orders of magnitude within one flow: a slowly moving liquid is in the incompressible regime, while gas, vapour pockets or detonation products may be transonic or carry strong shocks. Second, the phases exchange momentum, heat and mass on time scales from much slower to much faster than the flow, so that one computation contains frozen, finite-rate and equilibrium regions, and phase transition through metastable states.

Diffuse-interface models describe pure fluids, interfaces and mixtures with one set of equations throughout the domain [1, 74, 76]. They derive from the Baer–Nunziato seven-equation model [5], where each phase

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has its own velocity and pressure. Velocity relaxation gives the two-pressure six-equation model, additional pressure relaxation the five-equation model of Kapila et al. [2, 45, 60], and further thermal and chemical relaxation the four-equation pT model and the homogeneous equilibrium model [36, 56, 67]. Saurel, Petitpas and Berry [78] proposed computing interfaces and mixtures with the six-equation model and stiff pressure relaxation: its frozen sound speed is monotone in the volume fraction, its volume-fraction equation contains only the pressure work, and the mixture total energy, redundant for smooth flows, is added as a seventh, "+1" equation to recover the conservative mixture energy at shocks. Pelanti and Shyue [68] made the approach mixture-energy consistent and added heat and mass transfer [77, 87], and Pelanti [67] extended the relaxation procedures to arbitrary rates and to the Noble–Abel stiffened-gas [50] and Mie–Grüneisen equations of state.

A known difficulty of the six-equation model is its behaviour at shocks in mixtures: the model is non-conservative, its shock relations [75] are not those of the numerical shock layer, and the energy deposited there is shared between the phases in a scheme-dependent way. Saurel et al. [78] showed that the computed post-shock state then deviates from the five-equation solution for strong shocks, and corrected it by an artificial shock-layer heat exchange with a coefficient calibrated for each mixture [70]. A second difficulty is stiffness of two kinds: acoustic waves, fast compared with the flow at low Mach numbers, and mechanical, thermal and chemical relaxation, whose rates range from zero to infinity and may differ by orders of magnitude between processes. The methods above [26, 67, 68, 78] and the seven-equation solver of Sirianni and Re [81] are explicit Godunov-type schemes. They integrate the relaxation after a hyperbolic step restricted by the acoustic CFL condition. Since a small amount of liquid raises a mixture's frozen sound speed to nearly the liquid value, a slow flow is then advanced with the time step of the much faster acoustic waves. The upwind dissipation, which scales with the sound speed, also destroys the low-Mach asymptotics of the pressure [28, 37, 38]. Preconditioning corrects the scaling of the dissipation [66, 86] and, with implicit time integration, removes the time step restriction [53], at the price of large nonlinear systems coupling all variables, whose convergence is difficult to control. Semi-implicit methods avoid both difficulties by treating only the stiff terms implicitly and the nonlinear transport explicitly.

Pressure-based semi-implicit methods go back to Harlow and Welch [41] and Harlow and Amsden [40], with fundamental contributions by Casulli and Greenspan [17], Klein [47], Munz et al. [59] and Park and Munz [65]. Conservative, shock-capturing versions split the flux into a convective part, whose eigenvalues are the flow velocity or zero, and a pressure part carrying the acoustic waves [54, 85, 88]. Dumbser and Casulli [33] combined the Toro–Vázquez-Cendón splitting with an implicit pressure sub-system for general equations of state. This mildly nonlinear system, whose nonlinearity is confined to the diagonal through the equation of state, is solved by the nested Newton method of Casulli and Zanolli [18]. Higher-order and all-Mach finite volume and discontinuous Galerkin extensions followed [10, 12–14, 62, 82]. In ideal magnetohydrodynamics the Alfvén and magnetosonic waves were successively removed from the explicit step [15, 29, 32, 35], and Chen et al. [20] gave a pressure-based formulation for a general equation of state, enforcing pressure positivity inside the implicit solve. Implicit–explicit (IMEX) Runge–Kutta schemes [3, 64] provide second- and higher-order time integration; their stiffly accurate, L-stable members preserve relaxation limits [9, 11, 64]. A scheme is asymptotic preserving if, with a Mach-independent time step, its zero-Mach limit is a consistent discretisation of the incompressible equations [44, 46]; such all-speed schemes have been analysed for the Euler equations [12, 23, 27, 31, 39, 61]. These developments concern mainly single-phase flows.

For diffuse-interface two-phase models, pressure-based, all-Mach and IMEX schemes are still few: they exist for weakly compressible flows under a Baer–Nunziato type model [71], low-Mach flows with mass transfer [30], a two-phase model with phasic pressures [57], a two-fluid single-temperature model [55] and, with an all-regime Lagrange-projection scheme treating acoustics and phase transition implicitly, homogeneous two-phase models [19]. Convection–acoustics splittings have been derived for the Baer–Nunziato [84] and Kapila [34] models. Battisti and Boscheri [7] designed a linearly implicit, asymptotic preserving and well-balanced scheme for the seven-equation model, with explicit Rusanov convective fluxes, centrally discretised implicit pressure terms and stiff relaxation sources solved separately. Closest to the present work, Schropff, Petitpas and Daniel [80] derived a Zha–Bilgen type splitting of the single-velocity two-pressure model and a one-dimensional semi-implicit scheme solving the pressure sub-system by Picard iterations and correcting the phase internal energies to conserve total energy; their computations apply instantaneous pressure relaxation after the IMEX step. These schemes thus decouple the two stiffnesses, restricting the relaxation to instantaneous pressure equilibrium or solving it in a separate step. Outside

diffuse-interface models, semi-implicit two-fluid system codes such as RELAP5 [42] linearise interphase heat and mass transfer in time within the implicit pressure step and eliminate them, with the junction velocities, to leave one pressure equation per volume. Similarly, the all-Mach volume-of-fluid solver of Saade, Lohse and Fuster [73] solves pressure and temperature, with heat conduction, as one coupled system.

Here we treat both stiffnesses in a single implicit step. The pressure work and the three relaxation processes, with finite or infinite rates under the time-scale ordering of Pelanti [67], enter one local closure per cell. It reduces to a scalar equation in the cell pressure or, for finite mechanical relaxation, in the volume increment. The implicit step is a nonlinear system for the cell pressures, solved by Newton's method with a sparse three-point Jacobian, five-point in 2D. The cells are coupled through the face velocities as in the mildly nonlinear systems of [18, 33], while the nonlinearity from the equations of state and the relaxation is local to each cell. The closure uses the equations of state without linearisation, and each finite-rate relaxation is one backward-Euler step over the implicit step. One unified formula thus covers every rate, from the closure without that relaxation at zero rate to the equilibrium closure in the infinite-rate limit. The arbitrary-rate procedures of Pelanti [67] and of Sirianni and Re [81] integrate the relaxation equations with coefficients frozen over each step or sub-step and, in the infinite-rate limit, reach equilibrium only approximately or emulate it with large finite rates, respectively. Sirianni and Re also treat concurrent finite-rate mechanical and thermo-chemical relaxation, which the ordering used here excludes. The scheme is asymptotic preserving for both stiffnesses. At fixed mesh, under non-degeneracy assumptions and for data on the equilibrium manifold, it converges to the corresponding equilibrium scheme as each relaxation rate tends to infinity, and it reduces to the scheme without that relaxation as the rate vanishes. It is formally asymptotic preserving in the low Mach number limit, with a time step set only by a material-speed CFL condition, whatever the Mach number and relaxation rates.

The second ingredient concerns shocks: the difference between the conservative and non-conservative mixture internal energies, which at a shock is the scheme's dissipation, is distributed between the phases according to the model's shock relations [75], without a calibrated correction. We show that for weak shocks the share consistent with these relations is unique, compute for every shock strength the constant share for which the shock profile of the relaxed model ends on the Hugoniot of each phase, and distribute the remaining discretisation error in the unique linear proportion that keeps both phases admissible whenever possible.

The discretisation follows the Toro–Vázquez-Cendón splitting [85], with the mass and internal energy fluxes evaluated at the new velocity. The explicit step transports all quantities with the flow velocity by the path-conservative Rusanov scheme [16, 63] with material-speed dissipation. The implicit step carries the pressure, its work on each phase and the increments of the mass and internal energy fluxes from the old to the new velocity. Its central operator is supplemented by Laplace dissipation of pressure and velocity with acoustic coefficients: the upwind terms of the acoustic Riemann solution, weighted by the deficit of the backward-Euler viscosity relative to the upwind one, so that it vanishes at low Mach numbers. This dissipation is first order at local extrema and second order in smooth regions [43]. The stiffly accurate LSDIRK2 scheme [8] gives second order in time. The scheme applies, in one and two dimensions, to Mie–Grüneisen phases with instantaneous mechanical relaxation and to stiffened-gas and Noble–Abel stiffened-gas phases with all three relaxations. We prove that it preserves moving contacts between stiffened-gas or Noble–Abel stiffened-gas phases, at uniform temperature when thermal or chemical relaxation is active.

The scheme is validated on twenty-two one- and two-dimensional problems, counting the single- and two-phase Taylor–Green vortices separately: the convergence and low-Mach tests adapted from Battisti and Boscheri [7], single-phase Riemann problems at low and high Mach numbers and two low-Mach advection problems from Schropff et al. [80], and the interface, shock, cavitation, evaporation and depressurisation problems of [67, 68, 78].

The rest of the paper is organised as follows. Section 2 presents the model, its relaxed limits, the equations of state and the split form, Section 3 the scheme and the proofs of its properties, Section 4 the numerical results and Section 5 the conclusions.

2. System of governing equations

We consider the single-velocity six-equation two-phase model of Saurel, Petitpas and Berry [78], with the mixture total energy as additional equation and with the heat- and mass-transfer terms of Pelanti and Shyue [68] and Pelanti [67]. In vector form it reads

$$\begin{cases} \dfrac{\partial \alpha_1}{\partial t} + \mathbf{u}\cdot\nabla\alpha_1 = \mathcal{P} + \dfrac{\mathcal{M}}{\rho_I}, \\ \dfrac{\partial m_1}{\partial t} + \nabla\cdot(m_1\mathbf{u}) = \mathcal{M}, \\ \dfrac{\partial m_2}{\partial t} + \nabla\cdot(m_2\mathbf{u}) = -\mathcal{M}, \\ \dfrac{\partial (\rho\mathbf{u})}{\partial t} + \nabla\cdot\big(\rho\mathbf{u}\otimes\mathbf{u} + \bar{p}\,\mathbf{I}\big) = \mathbf{0}, \\ \dfrac{\partial \varepsilon_1}{\partial t} + \nabla\cdot(\varepsilon_1\mathbf{u}) + \alpha_1 p_1\,\nabla\cdot\mathbf{u} = -p_I\mathcal{P} + \mathcal{Q} + e_I\mathcal{M}, \\ \dfrac{\partial \varepsilon_2}{\partial t} + \nabla\cdot(\varepsilon_2\mathbf{u}) + \alpha_2 p_2\,\nabla\cdot\mathbf{u} = p_I\mathcal{P} - \mathcal{Q} - e_I\mathcal{M}, \\ \dfrac{\partial E}{\partial t} + \nabla\cdot\big((E+\bar{p})\mathbf{u}\big) = 0. \end{cases} \tag{1}$$

The first equation evolves the volume fraction of phase 1, the next two enforce the conservation of the mass of each phase, the fourth, a vector equation, governs the mixture momentum, the next two account for the internal energy of each phase, with its pressure work, and the last for the conservation of the mixture total energy. Here, for phases $k = 1, 2$, α_k is the volume fraction, with $\alpha_1 + \alpha_2 = 1$, ρ_k the density, $m_k = \alpha_k\rho_k$ the partial density, e_k the specific internal energy and $\varepsilon_k = \alpha_k\rho_k e_k$ the phase internal energy per unit mixture volume; p_k, T_k, c_k and g_k are the phase pressure, temperature, sound speed and Gibbs free energy. The mixture density is $\rho = m_1 + m_2$, $\mathbf{u} = [u, v]^\mathsf{T}$ is the velocity field, $\bar{p} = \alpha_1 p_1 + \alpha_2 p_2$ the mixture pressure, and E the total energy per unit volume, the sum of the phase internal energies and the kinetic energy,

$$E = \varepsilon_1 + \varepsilon_2 + \tfrac{1}{2}\rho|\mathbf{u}|^2. \tag{2}$$

The right-hand sides are the mechanical, thermal and chemical relaxation terms

$$\mathcal{P} = \mu(p_1 - p_2), \qquad \mathcal{Q} = \theta(T_2 - T_1), \qquad \mathcal{M} = \nu(g_2 - g_1), \\ p_I = \frac{Z_2 p_1 + Z_1 p_2}{Z_1 + Z_2}, \qquad Z_k = \rho_k c_k. \tag{3}$$

The three terms drive the phases towards equilibrium:

- mechanical relaxation $\mathcal{P}$: the volume fraction changes, with the associated interfacial work, until $p_1 = p_2$;
- thermal relaxation $\mathcal{Q}$: interphase heat exchange, which drives $T_1 = T_2$;
- chemical relaxation $\mathcal{M}$: mass transfer by evaporation or condensation, $\mathcal{M} > 0$ from phase 2 to phase 1, which drives $g_1 = g_2$.

The rate parameters $\mu, \theta, \nu \geq 0$ set the speed of the three processes and may be stiff. The total-energy equation is the conservative one and fixes the mixture internal energy at shocks [78]. The interfacial density ρ_I and the specific internal energy e_I of the transferred mass are taken, for definiteness, as the mixture values, $\rho_I = \rho$ and $e_I = (\varepsilon_1 + \varepsilon_2)/\rho$. We follow the ordering of time scales of Pelanti [67]: $\mu = \infty$ if $\theta > 0$ or $\nu > 0$, and $\theta = \infty$ if $\nu > 0$. Mass transfer therefore takes place at $p_1 = p_2 = p$ and $T_1 = T_2 = T$. There, by the Gibbs relation $m_k T_k\,\mathrm{D}\mathcal{S}_k = \mathrm{D}\varepsilon_k + p_k\,\mathrm{D}\alpha_k - h_k\,\mathrm{D}m_k$ of each phase, with $\mathcal{S}_k$ the specific entropy and $h_k = e_k + p_k/\rho_k = g_k + T_k\mathcal{S}_k$ the specific enthalpy, the contributions of ρ_I and e_I to the entropy production of the two phases cancel, and the mixture entropy production of the transfer

is $\mathcal{M}(g_2-g_1)/T = \nu(g_2-g_1)^2/T \geq 0$ for any ρ_I, e_I; for the same reason ρ_I and e_I do not enter the closures of Sec. 3.2.3.

An equation of state $p_k = p_k(\rho_k, e_k)$ is required for each phase to close the system. In this work we consider two families: the Noble–Abel stiffened-gas law of Sec. 2.1, for which the internal energy is affine in the pressure, and the Mie–Grüneisen laws with nonlinear reference curves of Sec. 2.2.

Letting $\mathbf{U} = (\alpha_1, m_1, m_2, \rho\mathbf{u}, \varepsilon_1, \varepsilon_2, E)^{\mathsf{T}}$ denote the vector of unknowns, system (1) can be written in compact form as

$$\partial_t \mathbf{U} + \nabla \cdot \boldsymbol{\mathcal{F}}(\mathbf{U}) + \mathbf{B}(\mathbf{U}) \cdot \nabla \mathbf{U} = \mathbf{S}(\mathbf{U}), \tag{4}$$

where $\boldsymbol{\mathcal{F}}$ collects the flux vectors, $\mathbf{B}$ the non-conservative products and $\mathbf{S}$ the relaxation terms. For one-dimensional flows aligned with the x-direction, (1) reads

$$\frac{\partial}{\partial t}\begin{bmatrix}\alpha_1\\ m_1\\ m_2\\ \rho u\\ \varepsilon_1\\ \varepsilon_2\\ E\end{bmatrix} + \frac{\partial}{\partial x}\begin{bmatrix}0\\ m_1 u\\ m_2 u\\ \rho u^2+\bar{p}\\ \varepsilon_1 u\\ \varepsilon_2 u\\ (E+\bar{p})u\end{bmatrix} + \begin{bmatrix}u\,\partial_x\alpha_1\\ 0\\ 0\\ 0\\ \alpha_1 p_1\,\partial_x u\\ \alpha_2 p_2\,\partial_x u\\ 0\end{bmatrix} = \begin{bmatrix}\mathcal{P}+\mathcal{M}/\rho_I\\ \mathcal{M}\\ -\mathcal{M}\\ 0\\ -p_I\mathcal{P}+\mathcal{Q}+e_I\mathcal{M}\\ p_I\mathcal{P}-\mathcal{Q}-e_I\mathcal{M}\\ 0\end{bmatrix}, \tag{5}$$

i.e. $\partial_t\mathbf{U} + \partial_x\mathbf{F}(\mathbf{U}) + \mathbf{B}(\mathbf{U})\,\partial_x\mathbf{U} = \mathbf{S}(\mathbf{U})$. Without relaxation terms the system has seven eigenvalues,

$$\lambda_{1,7} = u \mp c_f, \qquad \lambda_{2,\dots,6} = u, \tag{6}$$

with the phase sound speeds and the frozen sound speed of the mixture

$$c_k^2 = \frac{\partial p_k}{\partial \rho_k}\Big|_{e_k} + \frac{p_k}{\rho_k^2}\frac{\partial p_k}{\partial e_k}\Big|_{\rho_k}, \qquad c_f^2 = \sum_k Y_k c_k^2, \qquad Y_k = \frac{m_k}{\rho}. \tag{7}$$

A fully explicit scheme is therefore restricted by

$$\Delta t \leq C \min_{\Omega} \frac{\Delta x}{|u| + c_f}, \tag{8}$$

with a Courant number C of order one. This restriction becomes prohibitive when the flow is slow compared with the sound speed, and the relaxation terms add their own stiffness.

Relaxed models For $\mu \to \infty$ the model reduces to the five-equation pressure-equilibrium model of Kapila et al. [45]; adding $\theta \to \infty$ gives the four-equation pT model and adding $\nu \to \infty$ the homogeneous equilibrium model (pTg) [67]. Their sound speeds satisfy the subcharacteristic ordering [36]

$$c_{pTg} \leq c_{pT} \leq c_p \leq c_f, \qquad \frac{1}{\rho c_p^2} = \sum_k \frac{\alpha_k}{\rho_k c_k^2}. \tag{9}$$

2.1. Noble–Abel stiffened-gas equation of state

The Noble–Abel stiffened-gas (NASG) law [50] has the parameters γ, π, q, q', c_v and b; with the specific volume $v = 1/\rho$,

$$\begin{aligned} p(\rho, e) &= \frac{(\gamma-1)\rho(e-q)}{1-\rho b} - \gamma\pi, \\ v(p,T) = \frac{(\gamma-1)c_v T}{p+\pi} + b, &\qquad e(p,T) = c_v T\,\frac{p+\gamma\pi}{p+\pi} + q, \end{aligned} \tag{10}$$

with the Gibbs free energy $g = (\gamma c_v - q')T - c_v T \ln\left(T^\gamma/(p+\pi)^{\gamma-1}\right) + bp + q$ and the sound speed $c^2 = \gamma(p+\pi)/\big(\rho(1-\rho b)\big)$; $b = 0$ gives the stiffened-gas (SG) law and $b = \pi = q = 0$ the ideal gas.

The form used by the scheme is the internal energy of phase k in a cell, where it occupies the volume α_k with mass m_k, at pressure p,

$$\varepsilon_k(\alpha_k, m_k, p) = \frac{(p + \gamma_k \pi_k)(\alpha_k - m_k b_k)}{\gamma_k - 1} + m_k q_k, \tag{11}$$

which is affine in (α_k, m_k) at fixed p, affine in p at fixed (α_k, m_k), and explicitly invertible for α_k. The admissible set of a phase, $\{\alpha > 0,\ m > 0,\ \alpha - mb > 0,\ \varepsilon - mq - (\alpha - mb)\pi > 0\}$, is a convex cone. The saturation temperature $T_{\mathrm{sat}}(p)$ solves $g_1(p, T) = g_2(p, T)$ on the branch with $v_2 > v_1$; this branch ends at a pressure p_c, above which no two-phase equilibrium exists.

2.2. Mie–Grüneisen equations of state

Detonation products and liquid nitromethane are described by Mie–Grüneisen laws

$$p = \Gamma\rho\big(e - \varepsilon_r(\rho)\big) + p_r(\rho), \qquad c^2 = \frac{(\Gamma + 1)(p - p_r)}{\rho} + p_r'(\rho), \tag{12}$$

with constant Γ, whose reference curve is an isentrope, $\varepsilon_r' = p_r/\rho^2$. We use the Cochran–Chan law [22] and the JWL law [51],

$$\begin{aligned}
&\text{Cochran–Chan:} && p_r(\rho) = A_1 x^{E_1} - A_2 x^{E_2}, \qquad x = \frac{\rho}{\rho_0},\\
&&& \varepsilon_r(\rho) = \frac{1}{\rho}\Big(\frac{A_1 x^{E_1}}{E_1 - 1} - \frac{A_2 x^{E_2}}{E_2 - 1}\Big),\\
&\text{JWL:} && p_r(\rho) = A e^{-R_1\rho_0/\rho} + B e^{-R_2\rho_0/\rho},\\
&&& \varepsilon_r(\rho) = \frac{A}{R_1\rho_0} e^{-R_1\rho_0/\rho} + \frac{B}{R_2\rho_0} e^{-R_2\rho_0/\rho}.
\end{aligned} \tag{13}$$

In phase-volume form $\varepsilon_k = m_k\varepsilon_r(m_k/\alpha_k) + \alpha_k\big(p - p_r(m_k/\alpha_k)\big)/\Gamma_k$, which is nonlinear in α_k; admissibility is $c^2 > 0$. Mie–Grüneisen phases are used with instantaneous mechanical relaxation only.

2.3. Split form of the model

To handle the different time scales of the model we split the fluxes in the manner of Toro and Vázquez-Cendón [85], which allows a different time discretisation for each sub-system according to its stiffness. The terms of (5) are separated into the advective part, which transports the masses, the momentum and the kinetic energy with the flow velocity and contains the non-conservative advection $u\,\partial_x\alpha_1$ of the volume fraction, and the pressure–relaxation part, which contains the pressure flux, the internal energy fluxes of the phases with their pressure work $\mathbf{B}^p\partial_x u$, the enthalpy flux of the mixture and the relaxation terms. Explicitly,

$$\begin{aligned}
\mathbf{F}^c(\mathbf{U}) &= \begin{bmatrix} 0 \\ m_1 u \\ m_2 u \\ \rho u^2 \\ 0 \\ 0 \\ \frac{1}{2}\rho u^3 \end{bmatrix}, \quad \mathbf{B}^c(\mathbf{U})\,\partial_x\mathbf{U} = \begin{bmatrix} u\,\partial_x\alpha_1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},\\
\mathbf{F}^p(\mathbf{U}) &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ \bar{p} \\ \varepsilon_1 u \\ \varepsilon_2 u \\ \big(E - \frac{1}{2}\rho u^2 + \bar{p}\big)u \end{bmatrix}, \quad \mathbf{B}^p(\mathbf{U})\,\partial_x u = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \alpha_1 p_1 \\ \alpha_2 p_2 \\ 0 \end{bmatrix} \partial_x u,
\end{aligned} \tag{14}$$

and the one-dimensional system in split form reads

$$\partial_t \mathbf{U} + \partial_x \mathbf{F}^c(\mathbf{U}) + \mathbf{B}^c(\mathbf{U})\,\partial_x \mathbf{U} + \partial_x \mathbf{F}^p(\mathbf{U}) + \mathbf{B}^p(\mathbf{U})\,\partial_x u = \mathbf{S}(\mathbf{U}). \tag{15}$$

Since both sub-systems contain non-conservative products, their wave structure is given in the primitive variables

$$\mathbf{Q} = (\alpha_1, \rho_1, \rho_2, u, p_1, p_2, p)^\mathsf{T}, \tag{16}$$

where p is the mixture pressure of the total energy, $\sum_k \varepsilon_k(\alpha_k, m_k, p) = E - \frac{1}{2}\rho u^2$ with the phase energies at pressure p, as in (11). We write $\Gamma_k = \rho_k^{-1}\partial p_k/\partial e_k|_{\rho_k}$ for the Grüneisen coefficient and $\kappa_k = \partial p_k/\partial \rho_k|_{e_k} = c_k^2 - \Gamma_k p_k/\rho_k$.

1. The advective sub-system is

$$\partial_t \mathbf{U} + \partial_x \mathbf{F}^c(\mathbf{U}) + \mathbf{B}^c(\mathbf{U})\,\partial_x \mathbf{U} = \mathbf{0}. \tag{17}$$

In the primitive variables it reads

$$\partial_t \mathbf{Q} + \mathbf{A}^c(\mathbf{Q})\,\partial_x \mathbf{Q} = \mathbf{0},$$
$$\mathbf{A}^c(\mathbf{Q}) = \begin{bmatrix} u & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & u & 0 & \rho_1 & 0 & 0 & 0 \\ 0 & 0 & u & \rho_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & u & 0 & 0 & 0 \\ * & * & 0 & \rho_1(\kappa_1 - \Gamma_1 e_1) & 0 & 0 & 0 \\ * & 0 & * & \rho_2(\kappa_2 - \Gamma_2 e_2) & 0 & 0 & 0 \\ * & * & * & K^c & 0 & 0 & 0 \end{bmatrix}, \tag{18}$$
$$K^c = \frac{\sum_k \alpha_k \rho_k \kappa_k/\Gamma_k - \left(E - \frac{1}{2}\rho u^2\right)}{\sum_k \alpha_k/\Gamma_k},$$

since the volume fraction and the velocity are advected, the masses are transported at fixed phase internal energies ε_k, and the kinetic energy is transported with the mass, so that $E - \frac{1}{2}\rho u^2$ is constant and determines p at the advected volumes and transported masses; κ_k, Γ_k and e_k are evaluated at the phase state ρ_k, p_k in the rows of p_k and at ρ_k and the pressure p in K^c. The entries $*$ are proportional to u. For stiffened and Noble–Abel stiffened gases $\kappa_k - \Gamma_k e_k = \Gamma_k\big(b_k(p_k + \gamma_k\pi_k)/(\gamma_k - 1) - q_k\big)$ and $K^c = \sum_k m_k\big(b_k(p + \gamma_k\pi_k)/(\gamma_k - 1) - q_k\big)\big/\sum_k(\alpha_k - m_k b_k)/(\gamma_k - 1)$; both vanish for ideal and stiffened gases with $q_k = 0$. The last three columns of $\mathbf{A}^c$ vanish, and the eigenvalues are

$$\lambda^c_{1,\dots,4} = u, \qquad \lambda^c_{5,6,7} = 0. \tag{19}$$

One eigenvector of the eigenvalue u is missing, and the sub-system is weakly hyperbolic, as the pressureless gas dynamics it contains. The characteristic speeds are the flow velocity and zero. The volume fraction is transported only by this sub-system; in the pressure–relaxation sub-system it changes only through the relaxation terms.

2. The pressure–relaxation sub-system is

$$\partial_t \mathbf{U} + \partial_x \mathbf{F}^p(\mathbf{U}) + \mathbf{B}^p(\mathbf{U})\,\partial_x u = \mathbf{S}(\mathbf{U}), \tag{20}$$

with the relaxation terms $\mathbf{S}$ of (5). In the primitive variables it reads

$$\partial_t \mathbf{Q} + \mathbf{A}^p(\mathbf{Q})\,\partial_x \mathbf{Q} = \tilde{\mathbf{S}}(\mathbf{Q}),$$
$$\mathbf{A}^p(\mathbf{Q}) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ (p_1 - p_2)/\rho & 0 & 0 & 0 & \alpha_1/\rho & \alpha_2/\rho & 0 \\ * & * & 0 & \rho_1\Gamma_1 h_1 & u & 0 & 0 \\ * & 0 & * & \rho_2\Gamma_2 h_2 & 0 & u & 0 \\ * & * & * & K^p & 0 & 0 & u \end{bmatrix}, \tag{21}$$
$$K^p = \frac{E - \frac{1}{2}\rho u^2 + \bar{p}}{\sum_k \alpha_k/\Gamma_k},$$

with $\tilde{\mathbf{S}} = (\partial\mathbf{Q}/\partial\mathbf{U})\,\mathbf{S}$, the coefficients evaluated as in $\mathbf{A}^c$ and the entries $*$ again proportional to u. The fourth row is $\partial_x\bar{p}/\rho$, $\bar{p} = \alpha_1 p_1 + \alpha_2 p_2$; each phase keeps its volume and density, and its pressure changes by the transport of ε_k and by the pressure work, with the coefficient $\rho_k\Gamma_k h_k$ of $\partial_x u$, h_k the specific enthalpy; the internal energy $E - \frac{1}{2}\rho u^2$ receives the enthalpy flux of the mixture minus the work $u\,\partial_x\bar{p}$ on the kinetic energy. The first three rows vanish and the last column has only its diagonal entry, so the characteristic polynomial of $\mathbf{A}^p$ is $\lambda^3(\lambda-u)^2(\lambda^2 - u\lambda - c_\mathrm{h}^2)$ and the eigenvalues are

$$\lambda^p_{1,2,3} = 0, \qquad \lambda^p_{4,5} = u, \qquad \lambda^p_{6,7} = \tfrac{1}{2}\Big(u \pm \sqrt{u^2 + 4c_\mathrm{h}^2}\Big), \qquad c_\mathrm{h}^2 = \sum_k Y_k\Gamma_k h_k. \tag{22}$$

For $c_\mathrm{h}^2 > 0$, $\lambda^p_6 > 0 > \lambda^p_7$ whatever the flow velocity, as for the Euler equations [85]. For ideal and stiffened gases with $q_k = 0$, $\Gamma_k h_k = c_k^2$, and the pressure–relaxation sub-system carries the whole frozen sound speed. In general $c_\mathrm{h}^2 + \sum_k Y_k(\kappa_k - \Gamma_k e_k) = c_f^2$, the second part coming from the compression of the phases by the mass fluxes of the advective sub-system. Through h_k, c_h depends on the reference energies of the phases, q_k or ε_r: for the water of Sec. 4.3 in its initial state $c_\mathrm{h} = 430\,\mathrm{m/s}$ against $c_f = 1569\,\mathrm{m/s}$. Where $c_\mathrm{h}^2 < 0$, as in this liquid water below 275 K at 10^5 Pa and in the low-pressure initial state of the nitromethane of Sec. 4.2.3, $\lambda^p_{6,7}$ are complex for $u^2 < -4c_\mathrm{h}^2$ and the sub-system alone is not hyperbolic. It is never solved on its own but together with the incremental mass fluxes described below, so that the implicit step carries the whole sound speed, c_f or, with relaxation, that of the active closure.

In the low Mach number regime the acoustic coupling of the two sub-systems is fast compared with the transport, which introduces severe stiffness associated with the pressure terms, in addition to the stiffness of the relaxation rates. The advective sub-system is therefore treated explicitly, with a numerical dissipation proportional to the flow velocity, and the pressure–relaxation sub-system implicitly. As Chen et al. [20] evaluate the enthalpy flux as $h^n(\rho u)^{n+1}$, the internal energy fluxes are evaluated with the internal energies at t^n and the velocity at t^{n+1}, $\varepsilon_k^n u^{n+1}$, the internal energy flux of the mixture as their sum and the pressure fluxes as $\bar{p}^{\,n+1}$ and $\bar{p}^{\,n+1}u^{n+1}$. So that the implicit step contains the whole acoustic coupling, the mass fluxes are evaluated in the same way, $m_k^n u^{n+1}$; the momentum and the kinetic energy are carried with the new velocity by the incremental mass flux $\sum_k m_k(u^{n+1} - u^n)$. These fluxes are split into an old-velocity part and an increment, $Xu^{n+1} = Xu^n + X(u^{n+1} - u^n)$ for $X \in \{m_1, m_2, \varepsilon_1, \varepsilon_2\}$: the first part is evaluated in the explicit step together with the advective fluxes, so that the masses and the internal energies receive the same numerical dissipation, and the increments, which depend on the unknown velocity, are solved together with the pressure–relaxation sub-system in Sec. 3.2. Since only the flow velocity enters the explicit step, the time step depends on the flow velocity alone, independently of the sound speed and of the relaxation rates.

3. Semi-implicit numerical scheme

After the notation, Sec. 3.1 presents the explicit transport and Sec. 3.2 the implicit update of the pressure–relaxation sub-system with its dissipation, local closure, relaxation closures and pressure solver. The partition of the energy defect is analysed in Sec. 3.2.2, the well-balanced and asymptotic preserving properties in Secs. 3.3–3.4; second order in time is obtained in Sec. 3.5, and Sec. 3.6 collects the fully discrete scheme in two space dimensions. Fig. 1 summarises the split and one stage of the scheme.

The computational domain $\Omega = [x_\mathrm{min}, x_\mathrm{max}] \times [y_\mathrm{min}, y_\mathrm{max}]$ is covered by a uniform Cartesian grid of $N_x \times N_y$ control volumes with spacings $\Delta x = (x_\mathrm{max} - x_\mathrm{min})/N_x$ and $\Delta y = (y_\mathrm{max} - y_\mathrm{min})/N_y$. The cell (i, j) has centre $\mathbf{x}_{i,j} = (x_i, y_j)$, its faces are denoted by $(i \pm \frac{1}{2}, j)$ and $(i, j \pm \frac{1}{2})$ and its corners by $(i \pm \frac{1}{2}, j \pm \frac{1}{2})$. All variables are cell averages, defined at the cell centres for both sub-systems; time levels are denoted by superscripts and the face differences by $\delta_x[F]_{i,j} = F_{i+1/2,j} - F_{i-1/2,j}$, $\delta_y[G]_{i,j} = G_{i,j+1/2} - G_{i,j-1/2}$. Where the y-direction is analogous it is omitted.

3.1. Explicit transport

The explicit step advances the advective sub-system (17) together with the old-velocity part of the internal energy fluxes of Sec. 2.3, $\varepsilon_k u^n$ and $(E - \frac{1}{2}\rho u^2)u^n$, so that every quantity is transported with the

flow velocity by the flux $\mathbf{F}^u(\mathbf{U}) = u\,(0, m_1, m_2, \rho u, \varepsilon_1, \varepsilon_2, E)^\mathsf{T}$. The α_1 row, $u\,\partial_x\alpha_1$, is not the derivative of any flux, and the step is discretised by the path-conservative Rusanov scheme [16, 24, 63],

$$\begin{aligned} \mathbf{U}_i^* &= \mathbf{U}_i^n - \frac{\Delta t}{\Delta x}\Big(\mathbf{D}^-_{i+1/2} + \mathbf{D}^+_{i-1/2}\Big), \\ \mathbf{D}^-_{i+1/2} + \mathbf{D}^+_{i+1/2} &= \mathbf{F}^u(\mathbf{U}^R) - \mathbf{F}^u(\mathbf{U}^L) + \int_0^1 \mathbf{B}^c(\mathbf{\Psi})\,\partial_\ell \mathbf{\Psi}\,\mathrm{d}\ell, \end{aligned} \tag{23}$$

where $\mathbf{\Psi}(\ell)$, $\ell \in [0,1]$, is the path joining the two states of the face, and the Rusanov dissipation $\mp\frac{1}{2}a_{i+1/2}(\mathbf{U}^R - \mathbf{U}^L)$ enters $\mathbf{D}^\mp_{i+1/2}$, with

$$a_{i+1/2} = \max\big(|u^L_{i+1/2}|, |u^R_{i+1/2}|\big), \tag{24}$$

the larger flow speed of the two face states, or its maximum over all faces of the same direction, so that the dissipation is proportional to the flow velocity.

For the rows of m_k, ρu, ε_k and E the non-conservative part vanishes, the fluctuations are the jump of the flux whatever the path, and (23) is the conservative update with the Rusanov flux [72],

$$\begin{aligned} X^*_{i,j} = X^n_{i,j} &- \frac{\Delta t}{\Delta x}\,\delta_x\big[F^e[X]\big]_{i,j} - \frac{\Delta t}{\Delta y}\,\delta_y\big[G^e[X]\big]_{i,j}, \\ & X \in \{m_1, m_2, \varepsilon_1, \varepsilon_2, \rho u, \rho v, E\}, \end{aligned} \tag{25}$$

with

$$F^e_{i+1/2,j}[X] = \tfrac{1}{2}\big(X^L u^L + X^R u^R\big) - \tfrac{1}{2}a_{i+1/2,j}\big(X^R - X^L\big), \tag{26}$$

and G^e likewise with v. With the mean face velocity $\bar u_{i+1/2} = \frac{1}{2}(u^L + u^R)$ the speed (24) is $a_{i+1/2} = |\bar u_{i+1/2}| + \frac{1}{2}|u^R - u^L|$, and (26) is identically

$$F^e_{i+1/2}[X] = X^\uparrow\,\bar u_{i+1/2} - \tfrac{1}{2}\big(u^L - u^R\big)^+\big(X^R - X^L\big), \tag{27}$$

with $X^\uparrow = X^L$ for $\bar u_{i+1/2} \geq 0$, $X^\uparrow = X^R$ otherwise, and $(\cdot)^+ = \max(\cdot, 0)$. The first term transports X with the mean face velocity from its upwind side, the second is a dissipation acting only at compressive faces, $u^L > u^R$. With the maximum a_Ω of (24) over the faces of the same direction, the dissipation coefficient is $\frac{1}{2}(a_\Omega - |\bar u_{i+1/2}|) - \frac{1}{4}(u^R - u^L) \geq \frac{1}{2}(u^L - u^R)^+$. For the row of α_1 the path integral is taken between the cell values $\alpha_{1,i}$ and $\alpha_{1,i+1}$, also for second-order face values, along the linear path with the face velocity frozen at the mean $\bar u_{i+1/2}$, $\int_0^1 u\,\partial_\ell\alpha_1\,\mathrm{d}\ell = \bar u_{i+1/2}\,(\alpha_{1,i+1} - \alpha_{1,i})$, and the fluctuation is split between the two cells as in Saurel and Abgrall [74], with the Rusanov flux $F^e[\alpha_1]$ of the volume $\alpha_1 u$,

$$D^-_{i+1/2}[\alpha_1] = F^e_{i+1/2}[\alpha_1] - \alpha_{1,i}\,\bar u_{i+1/2}, \quad D^+_{i+1/2}[\alpha_1] = \alpha_{1,i+1}\,\bar u_{i+1/2} - F^e_{i+1/2}[\alpha_1]. \tag{28}$$

Their sum is the path integral, and the update of the cell reads

$$\begin{aligned} \alpha^*_{1,i,j} = \alpha^n_{1,i,j} &- \frac{\Delta t}{\Delta x}\Big(\delta_x\big[F^e[\alpha_1]\big]_{i,j} - \alpha^n_{1,i,j}\,\delta_x[\bar u]_{i,j}\Big) \\ &- \frac{\Delta t}{\Delta y}\Big(\delta_y\big[G^e[\alpha_1]\big]_{i,j} - \alpha^n_{1,i,j}\,\delta_y[\bar v]_{i,j}\Big), \qquad \alpha^*_2 = 1 - \alpha^*_1, \end{aligned} \tag{29}$$

the Rusanov difference of the volume flux minus the cell's own volume fraction $\alpha_{1,i}$ times the divergence of the face velocities, $\alpha_{1,i}$ being the volume fraction of the mass and energy that the conservative fluxes of the cell compress, so that the volume, the mass and the energy of a trace phase stay consistent where the velocity jumps across an interface. With first-order face values (29) is a convex combination of the neighbouring volume fractions when $C \leq \frac{1}{2}$, in a first-order step and in the first stage of the IMEX scheme of Sec. 3.5, so that $\alpha^*_1 \in (0,1)$. In the second stage the update base and the flux state of Sec. 3.5 differ, and

no such bound holds; there, as for second-order face values, admissibility is ensured by the step rejection of Sec. 3.5.

The face values are MUSCL reconstructions [52] with minmod. The volume-transported quantities $\mathcal{X}_V = (\alpha_1, m_1, m_2, \varepsilon_1, \varepsilon_2)$ share one limiter,

$$\begin{aligned} X^L_{i+1/2} = X_i + \tfrac{1}{2}\sigma_i, \quad X^R_{i+1/2} = X_{i+1} - \tfrac{1}{2}\sigma_{i+1}, \quad \sigma_i = \phi_i\, \tfrac{1}{2}(X_{i+1} - X_{i-1}), \\ \phi_i = \min_{X \in \mathcal{X}_V} \frac{\operatorname{minmod}(X_i - X_{i-1}, X_{i+1} - X_i)}{\frac{1}{2}(X_{i+1} - X_{i-1})} \in [0,1], \end{aligned} \tag{30}$$

so that linear relations between them, in particular $\varepsilon_k = \mathcal{C}_k(p)\alpha_k + \big(q_k - b_k\mathcal{C}_k(p)\big)m_k$, $\mathcal{C}_k(p) = (p+\gamma_k\pi_k)/(\gamma_k - 1)$, at uniform pressure, hold on the faces; quantities that are constant on the stencil to round-off are excluded from the minimum. For stiffened and Noble–Abel stiffened gases the admissible states of a phase form the cone $\alpha_k > 0$, $m_k > 0$, $L_k > 0$ for the two linear functionals $L^{(1)}_k = \alpha_k - m_k b_k$ and $L^{(2)}_k = \varepsilon_k - m_k q_k - (\alpha_k - m_k b_k)\pi_k$. Since all components share ϕ_i, such a functional of a face state is $L(X^{L,R}) = L_i \pm \frac{\phi_i}{4}(L_{i+1} - L_{i-1})$, which is not a convex combination of the cell values and is not bounded by the minmod factors of the individual components, e.g. at a strong pressure jump coinciding with a jump of the phase content. The shared factor is therefore limited further, for every stiffened or Noble–Abel stiffened phase with $L_i > 0$ and both functionals,

$$\phi_i \le \frac{2L_i}{|L_{i+1} - L_{i-1}|}, \qquad L \in \{L^{(1)}_k, L^{(2)}_k\},\ k = 1,2, \tag{31}$$

which keeps L on both faces above $\frac{1}{2}L_i$, the bound that minmod gives for a single non-negative monotone component. The velocity is reconstructed component-wise with its own minmod slope, and the face momentum and energy are assembled from the face values.

The time step is given by the CFL condition of the explicit transport,

$$\Delta t = C\left(\frac{\max_\Omega |u|}{\Delta x} + \frac{\max_\Omega |v|}{\Delta y}\right)^{-1}, \tag{32}$$

with $C = 0.2$, and the first steps are limited by the start-up rule of Sec. 3.5. A proof of stability under (32) is not available, as for the other semi-implicit schemes of this class; the computations of Sec. 4 use (32), those of Sec. 4.6 a smaller fixed time step, and the low-Mach and liquid–liquid shock tubes of Sec. 4.2.2 end within the start-up, in backward-Euler steps.

3.2. Pressure–relaxation sub-system

The pressure flux, the pressure work and the relaxation terms, together with the increments of the mass and internal energy fluxes, are treated implicitly and solved together, within a single step, as follows:

1. for given cell pressures, the face velocities give the incremental fluxes, the transported phase states, the strain of the pressure work and the conservative updates of momentum and total energy;
2. a local closure in every cell, containing the pressure work and the three relaxation processes, returns the cell pressure and the phase states consistent with them;
3. the pressure system, which requires the closure pressure to equal the cell pressure, is solved by Newton's method, each residual evaluation repeating items 1 and 2.

The implicit step has the length τ, $\tau = \Delta t$ in a first-order step, for which the semi-discrete form below is written, and $\tau = a_{ii}\Delta t$ in a stage of Sec. 3.5; Sec. 3.6 collects the stage.

Semi-discrete implicit step With the incremental fluxes of Sec. 2.3, the semi-discrete implicit step, (20) without the old-velocity part of its internal energy fluxes and with the incremental mass fluxes, is

$$\begin{aligned} X^{\mathrm{tr}} &= X^* - \Delta t\, \partial_x\big(X^{\mathrm{face}}(u^{n+1} - u^n)\big), \qquad X \in \{m_k, \varepsilon_k\}, \\ (\rho u)^{n+1} &= (\rho u)^* - \Delta t\, \partial_x \bar{p}^{\,n+1} - \Delta t\, \partial_x\big(u^{n+1}\hat{M}\big), \\ E^{n+1} &= E^* - \Delta t\, \partial_x\Big(\textstyle\sum_k \varepsilon_k^{\mathrm{face}}(u^{n+1} - u^n) + \frac{1}{2}(u^{n+1})^2\hat{M} + \bar{p}^{\,n+1}u^{n+1}\Big), \end{aligned} \tag{33}$$

where $\hat{M} = \sum_k m_k^{\mathrm{face}}(u^{n+1} - u^n)$ is the incremental mass flux and the face values are taken from the explicit state. The phase states at t^{n+1} follow from the transported state X^{tr}, completed by the volumes $\alpha_k^{\mathrm{tr}} = \alpha_k^*(1 - s)$ with the strain $s = \Delta t\, \partial_x u^{n+1}$ of the pressure work, and the conservative mixture internal energy $(\rho e)^C = E^{n+1} - \big((\rho u)^{n+1}\big)^2/(2\rho^{n+1})$ through the local closure of Sec. 3.2.2. The momentum and energy updates depend on $\bar{p}^{\,n+1}$, and the closure returns $\bar{p}^{\,n+1}$ as a function of s and $(\rho e)^C$, so the implicit step reduces to one equation per cell for the pressure,

$$G(\bar{p}) \equiv \bar{p}^{\mathrm{cl}}\big(s(\bar{p}), X^{\mathrm{tr}}(\bar{p}), (\rho e)^C(\bar{p});\ \mu, \theta, \nu\big) - \bar{p} = 0, \tag{34}$$

a nonlinear and relaxing generalisation of the pressure wave equation $p^{n+1} = p^{\mathrm{tr}} - \Delta t\, \rho c^2 \partial_x u^{n+1}$.

Derivation of the pressure wave equation The structure of (34) is seen by linearising it. To first order in Δt the closure gives the cell pressure as the pressure P^{tr} of the transported state minus the response of the mixture to the compression,

$$\bar{p}^{\,n+1} = P^{\mathrm{tr}} - K^W s, \qquad K^W = \rho c_p^2, \tag{35}$$

with the Wood bulk modulus of the mixture at mechanical equilibrium, which the activated closure replaces by its own stiffness. P^{tr} is the pressure p of (16) on X^{tr}; since X^{tr} compresses the volumes, masses and energies alike, it is to first order the pressure at t^n advected with the flow. The momentum update of (33), without the incremental term, provides the missing link,

$$u^{n+1} = u^* - \frac{\Delta t}{\rho}\, \partial_x \bar{p}^{\,n+1}, \tag{36}$$

and inserting it into (35) eliminates the velocity, giving an elliptic equation for the pressure,

$$\bar{p}^{\,n+1} - \Delta t^2 K^W\, \partial_x\Big(\frac{1}{\rho}\, \partial_x \bar{p}^{\,n+1}\Big) = P^{\mathrm{tr}} - \Delta t\, K^W\, \partial_x u^*. \tag{37}$$

This is the pressure equation of semi-implicit schemes for single-phase flows [13, 33] and of the seven-equation scheme of Battisti and Boscheri [7], with the stiffness of the two-phase mixture. The scheme does not use the linearisation: the closure evaluates the pressure response exactly for the equations of state and the relaxation rates, and the residual (34) also contains the incremental fluxes and the distribution of the energy defect.

3.2.1. Implicit dissipation

The implicit terms are discretised by central differences on the faces. The central operator does not damp the odd–even velocity mode, and the material-speed dissipation of (26) does not act on pressure jumps, so that, as in [13, 33], numerical dissipation is added to the implicit step by means of the Laplace operator

$$\begin{aligned}\mathbb{L}(\eta, q)_{i,j} = &\frac{1}{\Delta x^2}\Big(\eta_{i+1/2,j}(q_{i+1,j} - q_{i,j}) - \eta_{i-1/2,j}(q_{i,j} - q_{i-1,j})\Big)\\ &+ \frac{1}{\Delta y^2}\Big(\eta_{i,j+1/2}(q_{i,j+1} - q_{i,j}) - \eta_{i,j-1/2}(q_{i,j} - q_{i,j-1})\Big),\end{aligned} \tag{38}$$

applied to the pressure and to the velocity only, the acoustic characteristic variables: the volumes, partial densities and phase energies already carry the material-speed dissipation of the explicit step, and a dissipation proportional to the sound speed on them would smear contacts by a factor $1/M$ more than the material dissipation.

Face velocity and face pressure On the face $(i + \frac{1}{2}, j)$, with the cell pressures $P_{i,j}$, the unknowns of Sec. 3.2.4, $\rho^*_{i,j} = \sum_k m^*_{k,i,j}$, $u^*_{i,j} = (\rho u)^*_{i,j}/\rho^*_{i,j}$ and $Z_{i,j} = \rho^*_{i,j} c_{f,i,j}$, c_f the frozen sound speed (7) of X^*, the face velocity is the average of the cell velocities plus the pressure acceleration of the two adjacent half cells, the face pressure is the density-weighted average that conserves momentum, and each carries the diffusive

flux of the other acoustic variable:

$$
\begin{aligned}
u_f &= \bar{u}_f^* - \frac{\tau(P_{i+1,j} - P_{i,j})}{\bar{\rho}_f \Delta x} - \frac{\eta_f^P}{\Delta x}\big(P_{i+1,j} - P_{i,j}\big),\\
p_f &= \frac{\rho^*_{i+1,j} P_{i,j} + \rho^*_{i,j} P_{i+1,j}}{\rho^*_{i,j} + \rho^*_{i+1,j}} - \frac{\eta_f^u}{\Delta x}\, D_f^*,\\
\eta_f^P &= \frac{\varphi_f \psi_f^P\, \Delta x}{Z_{i,j} + Z_{i+1,j}}, \qquad \eta_f^u = \frac{\varphi_f \psi_f^u\, Z_{i,j} Z_{i+1,j}\, \Delta x}{Z_{i,j} + Z_{i+1,j}},
\end{aligned}
\tag{39}
$$

with $\bar{u}_f^* = \frac{1}{2}(u^*_{i,j} + u^*_{i+1,j})$, $\bar{\rho}_f = \frac{1}{2}(\rho^*_{i,j} + \rho^*_{i+1,j})$ and, in one dimension, $D_f^* = u^*_{i+1} - u^*_i$. Without the factors $\varphi_f \psi_f$ the two diffusive terms are the upwind terms of the acoustic Riemann solution. In the momentum update the face pressure contributes $\tau\, \mathbb{L}(\eta^u, u^*)$, a bulk viscosity with coefficient $\frac{1}{2}\rho c \Delta x$ for equal impedances, acting on the velocity of X^*; the divergence of the face velocity enters the strain, the transport and the energy flux, and in the pressure equation (37) it contributes $\tau K^W \mathbb{L}(\eta^P, P)$, with $K^W \eta^P = \frac{1}{2} c \Delta x$ for equal impedances and $c_p = c_f = c$. The pressure term is implicit; the velocity term is explicit, with the diffusion number $\frac{1}{2}\varphi_f \psi_f^u c\tau / \Delta x \le \frac{1}{8}$ by (42), so that neither restricts the time step (32). In two dimensions the normal velocity difference is replaced by Δx times the mean of the two vertex divergences of the face,

$$
\begin{aligned}
D^*_{i+1/2,j} &= \Delta x\, \tfrac{1}{2}\big(\mathrm{div}_{i+1/2,j+1/2} + \mathrm{div}_{i+1/2,j-1/2}\big),\\
\mathrm{div}_{i+1/2,j+1/2} &= \frac{u^*_{i+1,j} + u^*_{i+1,j+1} - u^*_{i,j} - u^*_{i,j+1}}{2\Delta x}\\
&\quad + \frac{v^*_{i,j+1} + v^*_{i+1,j+1} - v^*_{i,j} - v^*_{i+1,j}}{2\Delta y},
\end{aligned}
\tag{40}
$$

so that the velocity term acts on the divergence only and leaves divergence-free motion, such as vortices, untouched; for y-independent data it reduces to the one-dimensional difference.

The face sweeps $u_f \tau$ during the step, of which the explicit flux has already swept $\bar{u}_f^e \tau$, $\bar{u}_f^e = \bar{u}_{i+1/2}$ of (27); the incremental flux supplies the difference from the upwind side of u_f, with the reconstructed face values of (26) as donors,

$$
I_f[X] = X_f^L\big[(u_f)^+ - (\bar{u}_f^e)^+\big] + X_f^R\big[(u_f)^- - (\bar{u}_f^e)^-\big], \qquad X \in \{m_k, \varepsilon_k\},
\tag{41}
$$

with $(\cdot)^- = \min(\cdot, 0)$. Added to the explicit flux (27), the increment gives the flux of the step, $X_f^L (u_f)^+ + X_f^R (u_f)^- - d_f (X_f^R - X_f^L)$ with d_f the dissipation coefficient of the explicit flux given after (27): m_k and ε_k are transported by the face velocity u_f from its upwind side with the face values of the explicit state, $\varepsilon_k u_f$ being the internal energy flux of the pressure–relaxation sub-system, and the explicit step contributes their common dissipation.

Acoustic weight Backward Euler already damps acoustic waves: for linear acoustics, one backward-Euler step with (39) has, at low wave numbers, the effective viscosity $\frac{1}{2}c^2\tau + \varphi\, \frac{1}{2} c \Delta x = \frac{1}{2} c \Delta x (C_a + \varphi)$, $C_a = c\tau/\Delta x$. Requiring that each step carries at least the upwind viscosity $\frac{1}{2} c \Delta x$ on resolved acoustic waves, and no more, gives

$$
\varphi_f = \Big(1 - \frac{\tau \min(c_{f,i,j}, c_{f,i+1,j})}{\Delta x}\Big)^+.
\tag{42}
$$

For $c\tau \ge \Delta x$, in particular in the low Mach number regime, the Laplace terms vanish. The two stages of the second-order scheme of Sec. 3.5 together damp resolved acoustic waves only at higher order in Δt, so that there the Laplace terms supply φ times the upwind viscosity. The weight and the impedances use the frozen sound speed c_f, the acoustic speed (6) of the model without relaxation, which bounds the sound speed c_{eq} of the active closure, c_p, c_{pT} or c_{pTg} of (9) for instantaneous relaxation [36]; where $c_{\mathrm{eq}}\tau < \Delta x \le c_f \tau$ the weight vanishes and the relaxed waves are damped by the implicit step only. Where X^* is not admissible, $c_f = \infty$ and the face carries no Laplace terms.

Second-order factors The factors $\psi \in [0,1]$ evaluate the Laplace terms on the reconstructed face states, $\psi\Delta_f = X^R - X^L$ for the face jump $\Delta_f = X_{i+1,j} - X_{i,j}$. The pressure is a function of the volume-transported quantities and the velocity, whose reconstruction is scaled by $\phi_s = \min(\hat\phi_{i,j}, \hat\phi_{i+1,j})$, with $\hat\phi$ the minimum of the shared factor (30) after the cap (31) and of the smoothness ratios, minmod over central slope, of the velocity components; with the minmod slope of P this gives

$$\begin{aligned} \psi_f^P &= 1 - \phi_s\big(1 - \psi_f^{\mathrm{mm}}[P]\big), \qquad \psi_f^u = \psi_f^{\mathrm{mm}}[u], \\ \psi_f^{\mathrm{mm}}[X] &= 1 - \frac{\sigma_{i,j}^{\mathrm{mm}}[X] + \sigma_{i+1,j}^{\mathrm{mm}}[X]}{2\Delta_f}, \end{aligned} \tag{43}$$

ψ_f^u being taken on the normal velocity component alone, also in two dimensions, where it multiplies D_f^*. The slopes of P are those of the Newton initial guess of the stage, the pressure of the previous stage, and those of u are taken on u^*, so that the factors are frozen during the solution of the stage. With unlimited central slopes, $\hat\phi = 1$ without the cap (31), σ are the central slopes and ψ is clipped to $[0,1]$. The pressure term is first order where P, a volume-transported quantity or a velocity component has a local extremum, in particular at contacts, and the velocity term where the normal velocity has one; both are second order where the solution is smooth, which is the local-extremum-diminishing criterion of Jameson [43].

3.2.2. Local closure

Backward Euler of the implicit part of (1), in which the volume fractions change only by relaxation and phase k does the pressure work $\alpha_k^* p_k\, s$, gives, with the closure input α_k^{tr}, $\chi_1 = +1$ and $\chi_2 = -1$,

$$\begin{aligned} \alpha_k &= \alpha_k^{\mathrm{tr}} + \alpha_k^* s + \chi_k R, \qquad R = \tau\big(\mu(p_1 - p_2) + \mathcal{M}/\rho_I\big), \\ m_k &= m_k^{\mathrm{tr}} + \chi_k \tau \mathcal{M}, \\ \varepsilon_k &= \varepsilon_k^{\mathrm{tr}} - \alpha_k^* p_k\, s - \chi_k\, \tau p_I \mu (p_1 - p_2) + \chi_k \tau\big(\mathcal{Q} + e_I \mathcal{M}\big) + \Lambda_k(\delta), \end{aligned} \tag{44}$$

so $\sum_k \alpha_k = 1$ for every pressure, since $\alpha_k^{\mathrm{tr}} + \alpha_k^* s = \alpha_k^*$. The last term distributes the defect between the non-conservative and the conservative mixture internal energy,

$$\delta = (\rho e)^C - \sum_k \varepsilon_k^{\mathrm{tr}} + s \sum_k \alpha_k^* p_k, \qquad \sum_k \Lambda_k(\delta) = \delta. \tag{45}$$

By (44) $\sum_k \varepsilon_k = (\rho e)^C$ for every Λ_k with $\sum_k \Lambda_k = \delta$, so the closed state is mixture-energy consistent without a final projection, and δ does not depend on the relaxation terms, whose contributions cancel between the phases. In smooth flow δ is a discretisation error; at a shock it is the dissipation of the scheme, the kinetic energy that its numerical viscosity converts into internal energy. Its partition between the phases is not fixed by the inviscid model and determines the limit shock relations of the scheme.

Shock relations and the weak-shock share We require each phase to follow its own Hugoniot [75, 78] from the pre-shock state a to the post-shock state b, $p_a < p_b$,

$$e_k - e_{k,a} + p_m (v_k - v_{k,a}) = 0, \qquad p_m = \tfrac{1}{2}(p_a + p_b), \qquad k = 1, 2, \tag{46}$$

the shock relations also used for the exact solutions of Sec. 4, v denoting the specific volume. Consider a stationary shock profile of the relaxed model with mass flux j, viscous stress Π, the Rayleigh line $p - \Pi = p_a + j^2(v_a - v)$ and the mixture energy balance $\mathrm{d}e = -(p - \Pi)\,\mathrm{d}v$. Each phase receives the heat $\mathrm{d}\mathcal{D}_k$,

$$\mathrm{d}e_k = -p\,\mathrm{d}v_k + \mathrm{d}\mathcal{D}_k, \quad Y_k\,\mathrm{d}\mathcal{D}_k = \lambda_k\,\mathrm{d}\mathcal{D}, \quad \mathrm{d}\mathcal{D} = \Pi\,\mathrm{d}v \geq 0, \quad \textstyle\sum_k \lambda_k = 1, \tag{47}$$

with $\lambda_k = \partial\Lambda_k/\partial\delta$ the local share. Summing over k recovers $\mathrm{d}e = -(p - \Pi)\,\mathrm{d}v$: whatever the share, the mixture ends on its Hugoniot, and the share only decides the end volumes of the phases. Integration of $\mathrm{d}e_k = -p\,\mathrm{d}v_k + \mathrm{d}\mathcal{D}_k$ over the profile and comparison with (46) show that the end state satisfies (46) if and only if

$$\mathcal{D}_k = \int_{\text{profile}} (p - p_m)\,\mathrm{d}v_k, \qquad k = 1, 2, \tag{48}$$

the heat received by phase k is its Hugoniot work minus its compression work along the profile.

Theorem 1 (Weak-shock share). *Let $\Delta p = p_b - p_a \to 0$, $K_k = \rho_k c_k^2$ and $\mathcal{G}_k = \frac{1}{2}\big(1 + \partial K_k/\partial p|_{\mathcal{S}_k}\big)$ the fundamental derivative of phase k, with α_k, K_k and $\mathcal{G}_k$ at the pre-shock state. Under (46) the heat received by phase k per unit mixture volume is*

$$Y_k \rho\, \mathcal{D}_k = \frac{\alpha_k \mathcal{G}_k}{6K_k^2}\,\Delta p^3 + O(\Delta p^4), \tag{49}$$

$\mathcal{D}_k$ being the entropy production $T_k \Delta \mathcal{S}_k$ of a single-phase shock of the same strength in phase k. Hence a share λ_k that is consistent with (46) for weak shocks is unique and equal to $w_k = \big(\alpha_k \mathcal{G}_k/K_k^2\big)/\sum_j \alpha_j \mathcal{G}_j/K_j^2$; for Noble–Abel stiffened gases $\mathcal{G}_k = (\gamma_k+1)/\big(2(1-\rho_k b_k)\big)$.

Proof. To the order required each phase is isentropic along a weak profile, and with $x = p - p_a$

$$v_k(p) = v_{k,a}\Big[1 - \frac{x}{K_k} + \mathcal{G}_k \frac{x^2}{K_k^2} + O(x^3)\Big], \quad \mathrm{d}v_k = v_{k,a}\Big[-\frac{1}{K_k} + 2\mathcal{G}_k \frac{x}{K_k^2} + O(x^2)\Big]\mathrm{d}x.$$

Inserting this in (48) with $p - p_m = x - \frac{1}{2}\Delta p$,

$$\begin{aligned}\mathcal{D}_k &= v_{k,a}\int_0^{\Delta p}\Big(x - \frac{\Delta p}{2}\Big)\Big[-\frac{1}{K_k} + 2\mathcal{G}_k\frac{x}{K_k^2}\Big]\mathrm{d}x + O(\Delta p^4)\\ &= v_{k,a}\,\frac{2\mathcal{G}_k}{K_k^2}\,\frac{\Delta p^3}{12} + O(\Delta p^4),\end{aligned}$$

since the first term integrates to zero; with $\rho Y_k v_{k,a} = \alpha_k$ this is (49). It equals $T_k \Delta \mathcal{S}_k$ of the single-phase Hugoniot of phase k at the same Δp [48], which does not depend on the path. If λ_k is a continuous function of the state, then on a weak profile $\lambda_k = \lambda_k(\mathbf{U}_a) + O(\Delta p)$, and (48) with (49) requires $\lambda_k(\mathbf{U}_a) = w_k(\mathbf{U}_a)$.

A share by mass, by $\alpha_k/(\gamma_k - 1)$ or by the volume-change fraction α_k/K_k is therefore not consistent with (46). For strong shocks the heat (48) depends on the profile, so that no closed-form gap between the end states reproduces it. Since the scheme distributes δ with a share that is constant while the shock crosses a cell, the share it needs is the constant share for which the profile (47) ends at the state (46). This share is computed directly. The end volume of the mixture, $v_b = \sum_k Y_k v_{k,b}$, with $v_{k,b}$ the volume of phase k at p_b on its Hugoniot (46), does not depend on the share, so that $j^2 = (p_b - p_a)/(v_a - v_b)$ and the profile is integrated over the fixed interval from v_a to v_b. With $\lambda_2 = 1 - \lambda_1$, (47) and $\sum_k Y_k v_k = v$ give

$$\frac{\mathrm{d}p}{\mathrm{d}v} = \frac{\sum_k Y_k r_k/e_{v,k} - 1}{\sum_k Y_k e_{p,k}/e_{v,k}}, \qquad \frac{\mathrm{d}v_1}{\mathrm{d}v} = \frac{r_1 - e_{p,1}\,\mathrm{d}p/\mathrm{d}v}{e_{v,1}}, \qquad \frac{\mathrm{d}\mathcal{D}}{\mathrm{d}v} = \Pi(v), \tag{50}$$

with $\Pi = p - p_a - j^2(v_a - v)$ by the Rayleigh line, $e_{p,k} = \partial e_k/\partial p|_v$, $e_{v,k} = \partial e_k/\partial v|_p + p$ and $r_k = (\lambda_k/Y_k)\Pi$; for SG and NASG phases $e_{p,k} = (v_k - b_k)/(\gamma_k - 1)$, $e_{v,k} = \gamma_k(p + \pi_k)/(\gamma_k - 1)$, and for Mie–Grüneisen phases $e_{p,k} = v_k/\Gamma_k$, $e_{v,k} = (p - p_r)(1 + 1/\Gamma_k) + \rho_k p_r'/\Gamma_k$. The share λ_1 is the root of $v_1(v_b; \lambda_1) = v_{1,b}$, which is increasing in λ_1; it is found by the Illinois method in $\mathrm{logit}(\lambda_1)$, and (50) is integrated by an adaptive Runge–Kutta–Fehlberg method with the Cash–Karp coefficients and relative tolerance 10^{-8}. The same integration gives the mixture dissipation $\mathcal{D} = \int \Pi\,\mathrm{d}v$ per unit mass, and the physical part of the dissipation of phase k per unit volume is

$$H_k = \lambda_k\,\rho\,\mathcal{D}, \qquad H = \sum_k H_k = \rho\,\mathcal{D}. \tag{51}$$

For weak shocks, $p_b - p_a < 10^{-2}\min_k K_k$, the residual of the shooting is $O(\Delta p^3)$ and the limit of Theorem 1 is used with $\alpha_k = \rho Y_k v_{k,a}$, $\lambda_k = w_k$ and $\mathcal{D} = \sum_k Y_k v_{k,a} \mathcal{G}_k \Delta p^3/(6K_k^2)$.

The pre- and post-shock states are taken on the compact stencil of the cell, the cell and its neighbours, a five-point cross in 2D: $p_a = p_{\mathrm{lo}} < p_b = p_{\mathrm{hi}}$ are the extrema of the mixture-equilibrium pressure of X^* on the

stencil, the pre-shock phase volumes $v_{k,a} = \alpha_k/m_k$ are those of its lowest-pressure cell, Y_k and ρ are those of the cell itself, and H_k is computed once per stage and frozen. A positive defect is distributed with the share $\lambda_k = H_k/H$ up to H, while the part of the defect of a stage above H, which exceeds the dissipation of the whole shock, and a negative defect are discretisation errors. The amount deposited is not set by H but is the defect produced by the scheme, which the conservation of the mixture total energy fixes over the profile; the budget only selects its share. Each cell evaluates the share on the pressure range of its own stencil, the whole jump when the profile spans at most three cells and a part of it otherwise, and since the share varies slowly with the shock strength and tends to w_k for weak shocks, the cells of a smeared profile use nearly the same share. H_k is computed only in cells where the central differences of u^* and v^* have negative sum and the stencil pressure varies. By the Bethe–Weyl theorem [58] every shock satisfying the entropy condition is compressive where $\mathcal{G}_k > 0$, which holds for SG and NASG phases and for the Mie–Grüneisen phases of Sec. 4 over the range of states of the tests, so that these cells contain every admissible shock. Elsewhere, and where no admissible end state or profile exists, $H_k = 0$.

Partition of the discretisation error Let $\mathcal{E}_k$ be the internal energy of phase k above its admissibility bound, for SG and NASG phases

$$\mathcal{E}_k = \varepsilon_k^{\mathrm{tr}} - m_k^{\mathrm{tr}} q_k - (\alpha_k^{\mathrm{tr}} - m_k^{\mathrm{tr}} b_k)\pi_k = \frac{(p_k + \pi_k)(\alpha_k - m_k b_k)}{\gamma_k - 1}\Big|_{\mathrm{tr}}, \tag{52}$$

and for Mie–Grüneisen phases the thermal energy above the cold curve, $\varepsilon_k - m_k \varepsilon_r$.

Proposition 2 (Partition of the error). *Let both phases be admissible, $\mathcal{E}_k > 0$, and let the error be distributed at fixed phase volumes linearly, $\Lambda_k = \lambda_k \delta$ with $\sum_k \lambda_k = 1$. Both phases remain admissible, $\mathcal{E}_k + \lambda_k \delta > 0$, for every $\delta \in (-\sum_j \mathcal{E}_j, 0)$ if and only if*

$$\lambda_k = \lambda_k^{\mathcal{E}} = \frac{\mathcal{E}_k}{\sum_j \mathcal{E}_j}, \tag{53}$$

and then both phases remain admissible for every $\delta > -\sum_j \mathcal{E}_j$.

Proof. Let $S = \sum_j \mathcal{E}_j$. If $\mathcal{E}_k + \lambda_k \delta > 0$ for all $\delta \in (-S, 0)$, letting $\delta \to -S$ gives $\lambda_k \le \mathcal{E}_k/S$ for $k = 1, 2$. Summing, $1 = \sum_k \lambda_k \le \sum_k \mathcal{E}_k/S = 1$, so every inequality is an equality. Conversely, $\mathcal{E}_k + \lambda_k^{\mathcal{E}} \delta = \mathcal{E}_k(1 + \delta/S) > 0$ for every $\delta > -S$. For $\delta \le -S$ no admissible partition exists, since $\sum_k (\mathcal{E}_k + \Lambda_k) = S + \delta \le 0$.

The share $\lambda_k^{\mathcal{E}}$ changes the energy of both phases above their bounds by the same relative amount: for SG and NASG phases at fixed volumes, $p_k + \pi_k$ is multiplied by $1 + \delta/\sum_j \mathcal{E}_j$. The proposition is stated at fixed phase volumes. In the closure the volumes change by $\alpha_k - \alpha_k^{\mathrm{tr}}$, with $\sum_k (\alpha_k - \alpha_k^{\mathrm{tr}}) = s$, and the margin of an SG or NASG phase at the solution is $\mathcal{E}_k + \Lambda_k(\delta) - (p + \pi_k)(\alpha_k - \alpha_k^{\mathrm{tr}})$; the share uses $\mathcal{E}_k$ of the transported state, and admissibility at the solution is checked by the closure. When both phases are at their bounds, $\sum_j \mathcal{E}_j = 0$, the error is distributed by mass. For Mie–Grüneisen phases $\mathcal{E}_k > 0$ is the positivity of the thermal energy only; their admissibility $c_k^2 > 0$ is checked at the solution of the closure and ensured by the step rejection of Sec. 3.5. Altogether

$$\Lambda_k(\delta) = \begin{cases} \min(\delta, H)\,\dfrac{H_k}{H} + \big(\delta - \min(\delta, H)\big)\lambda_k^{\mathcal{E}}, & \delta \ge 0, \\ \delta\,\lambda_k^{\mathcal{E}}, & \delta < 0, \end{cases} \qquad H = \sum_k H_k, \tag{54}$$

a continuous, piecewise affine function of δ with break points $\delta = 0$ and $\delta = H$. On the active piece $\Lambda_k = \lambda_k \delta + \xi_k$, with $\xi_k = H_k - H\lambda_k^{\mathcal{E}}$ on the third piece and $\xi_k = 0$ otherwise. With a common pressure $\delta = D_0 + ps$, $D_0 = (\rho e)^C - \sum_k \varepsilon_k^{\mathrm{tr}}$. The NASG p closure of Sec. 3.2.3 selects the active piece at every trial pressure of its root solve, so that its output is continuous in P; the other closures that use Λ_k, those with finite μ or θ and those of Mie–Grüneisen phases, select it at the current Newton iterate P, and their output jumps where $\delta(P)$ crosses a break point, by an amount that vanishes at a solution of the closures with a common pressure and is of order $sR(p_1 - p_2)$ for finite μ.

3.2.3. Relaxation closures

The relaxation terms are not split from the transport: they enter the closure (44), which for each set of rates reduces to one scalar equation per cell, $\Phi(p) = 0$ in the cell pressure or, for finite μ, $\mathcal{F}(R) = 0$ in the volume increment. The closure returns the phase states and the mixture pressure $\bar{p}^{\mathrm{cl}} = \sum_k \alpha_k p_k$. The finite-rate closures are written with $\zeta = 1/(\tau\mu)$, $1/(\tau\theta)$ or $1/(\tau\nu)$ in $[0, \infty]$; $\zeta = \infty$ switches the process off and $\zeta = 0$ gives exactly the corresponding equilibrium closure, for ν where $p < p_c$, so that one formula covers every rate. For infinite rates the equilibrium closures are used directly. The equation $\Phi(p) = 0$ of the pT, finite-θ, finite-ν and pTg closures is solved by Brent's method in $\ln(p - \max_k(-\pi_k))$.

Instantaneous mechanical relaxation: p closure, $\mu = \infty$, $\theta = \nu = 0$ With $p_1 = p_2 = p_I = p$ and $\alpha_k^* s + \chi_k R = \alpha_k - \alpha_k^{\mathrm{tr}}$, (44) becomes the discrete Gibbs relation

$$\varepsilon_k = \varepsilon_k^{\mathrm{tr}} - p\,(\alpha_k - \alpha_k^{\mathrm{tr}}) + \Lambda_k(\delta), \qquad \delta = D_0 + ps, \tag{55}$$

independent of α_k^* and p_I. By (11), on the active piece of (54),

$$\begin{aligned} \alpha_k(p) &= \frac{(\gamma_k - 1)\big[A_k + B_k p\big] + (p + \gamma_k \pi_k) m_k b_k}{\gamma_k (p + \pi_k)}, \\ A_k &= \varepsilon_k^{\mathrm{tr}} - m_k q_k + \lambda_k D_0 + \xi_k, \qquad B_k = \alpha_k^{\mathrm{tr}} + \lambda_k s, \end{aligned} \tag{56}$$

and $\Phi(p) = \sum_k \alpha_k(p) - 1$. Here

$$\alpha_k'(p) = -\frac{(\gamma_k - 1) N_k}{\gamma_k (p + \pi_k)^2}, \qquad N_k = A_k - (B_k - m_k b_k)\pi_k.$$

By (52), $N_k = \mathcal{E}_k + \lambda_k(D_0 - s\pi_k) + \xi_k$, with $\mathcal{E}_k > 0$ if the transported state of phase k is admissible. As $p \downarrow \max_k(-\pi_k)$, $\Phi \to +\infty$ if $N_k > 0$ for the phases with the smallest π_k; as $p \to \infty$, $\Phi \to \sum_k (m_k b_k - B_k)/\gamma_k$, since $\sum_k B_k = 1 - s + s = 1$, and this limit is negative for $s \geq 0$. Under these two conditions a root exists; it is unique when $N_k > 0$ for both phases, in which case Φ is decreasing. The root is bracketed on $(\max_k(-\pi_k), \infty)$ and found by a Newton iteration safeguarded by bisection, and admissibility is checked at the solution. For a Mie–Grüneisen phase (55) with (12) gives $\alpha_k(p)$ as the root of

$$f_k(\alpha) = m_k \varepsilon_r(m_k/\alpha) + \alpha\,\frac{p - p_r(m_k/\alpha)}{\Gamma_k} + p\,\alpha - A_k - B_k p = 0, \quad f_k'(\alpha) = \frac{\rho_k c_k^2}{\Gamma_k},$$

with $A_k = \varepsilon_k^{\mathrm{tr}} + \lambda_k D_0 + \xi_k$, unique where $c_k^2 > 0$ on the bracket. The bracket is searched on the branch $c_k^2 > 0$, on which f_k is increasing: an initial volume with $c_k^2 \leq 0$, as for a trace phase that has received mass and energy without volume, is first doubled until $c_k^2 > 0$. The root is found by a Newton iteration in α safeguarded by bisection in $\ln \alpha$. A stiffened or Noble–Abel stiffened partner phase is admitted only where $N_k > 0$, and the outer equation is solved by the same safeguarded Newton iteration, without a global monotonicity result.

Finite mechanical relaxation: $\mu < \infty$, $\theta = \nu = 0$ Since $\alpha_k^{\mathrm{tr}} + \alpha_k^* s = \alpha_k^*$, $\alpha_1 = \alpha_1^* + R$, $\alpha_2 = \alpha_2^* - R$, and p_I is given by (3) with Z_k frozen on X^*. For given R, (44) with (11) is the 2×2 linear system

$$\begin{aligned} &\Big(\frac{\alpha_k - m_k b_k}{\gamma_k - 1} + \alpha_k^* s\Big) p_k + \chi_k R\, p_I - \lambda_k s\,(\alpha_1^* p_1 + \alpha_2^* p_2) \\ &\qquad = \varepsilon_k^{\mathrm{tr}} - m_k q_k - \frac{\gamma_k \pi_k (\alpha_k - m_k b_k)}{\gamma_k - 1} + \lambda_k D_0 + \xi_k \end{aligned} \tag{57}$$

for (p_1, p_2), $k = 1, 2$, with $\delta = D_0 + s \sum_k \alpha_k^* p_k$. R solves the scalar equation

$$\mathcal{F}(R) = \zeta R - \big(p_1(R) - p_2(R)\big) = 0.$$

To first order in R and s, increasing R decreases p_1 and increases p_2 where $(\gamma_k - 1)p_I + p_k + \gamma_k \pi_k > 0$, so that $\mathcal{F}$ is increasing; relaxation reduces $|p_1 - p_2|$, and the root lies between 0 and $\bar{R} = (p_1 - p_2)|_{R=0}/\zeta$. If $\bar{R}$ is not admissible, $\mathcal{F}$ tends to infinity with the correct sign at the singularity of (57), and bisection towards 0 gives an admissible bracket, on which $\mathcal{F} = 0$ is solved by Brent's method. For $\zeta = 0$, $p_1 = p_2 = p_I$ and (57) is the p closure (55). For $\mu = 0$, $R = 0$: each phase does the Gibbs work at its own pressure, $\varepsilon_k - \varepsilon_k^{\mathrm{tr}} = -p_k(\alpha_k - \alpha_k^{\mathrm{tr}}) + \Lambda_k$, and $\alpha_k = \alpha_k^{\mathrm{tr}}/(1 - s) = \alpha_k^*$.

Instantaneous thermal relaxation: pT closure, $\mu = \theta = \infty$, $\nu = 0$ The masses are $m_k = m_k^{\mathrm{tr}}$. The mixture energy $\sum_k m_k e_k(p,T) = (\rho e)^C$ gives the temperature explicitly, and

$$T(p) = \frac{(\rho e)^C - \sum_k m_k q_k}{\sum_k m_k c_{vk} \frac{p+\gamma_k \pi_k}{p+\pi_k}}, \qquad \Phi(p) = \sum_k m_k v_k\big(p, T(p)\big) - 1, \tag{58}$$

independent of α_k^{tr}, s and Λ_k: the volume fractions are dependent variables, as in the four-equation pT model. The admissible root is unique [68]; the closure returns $\alpha_k = m_k v_k$, $\varepsilon_k = m_k e_k$ and $\bar{p}^{\mathrm{cl}} = p$.

Finite thermal relaxation: $\mu = \infty$, $\theta < \infty$, $\nu = 0$ In (55) ε_1 gains and ε_2 loses the heat $\tau\mathcal{Q} = \tau\theta(T_2 - T_1)$. By (56), α_k is linear in $\tau\mathcal{Q}$, $\alpha_k = \alpha_k^0(p) + \chi_k(\gamma_k - 1)\tau\mathcal{Q}/\big(\gamma_k(p+\pi_k)\big)$, with α_k^0 the volumes of the p closure, and the temperatures $T_k = (\alpha_k/m_k - b_k)(p+\pi_k)/\big((\gamma_k - 1)c_{vk}\big)$ become

$$T_k = T_k^0(p) + \chi_k \frac{\tau\mathcal{Q}}{C_{pk}}, \qquad C_{pk} = \gamma_k c_{vk} m_k,$$

the heat at constant pressure being the enthalpy increase. Hence, in closed form,

$$\tau\mathcal{Q}(p) = \frac{T_2^0(p) - T_1^0(p)}{C_{p1}^{-1} + C_{p2}^{-1} + \zeta}, \tag{59}$$

and $\Phi(p) = \sum_k \alpha_k(p) - 1$. For $0 < \theta < \infty$ the closure requires $N_k > 0$ for both phases; $\theta = 0$ gives the solution of the p closure. For $\zeta = 0$, $T_1 = T_2$ at every p, and the root of Φ is that of the pT closure, where $\sum_k \varepsilon_k = (\rho e)^C$.

Finite chemical relaxation: $\mu = \theta = \infty$, $\nu < \infty$ The density $\rho = \sum_k m_k^{\mathrm{tr}}$ is fixed. For given p and vapour mass fraction $Y = Y_2$,

$$\begin{aligned} T(p,Y) &= \frac{(\rho e)^C/\rho - (1-Y)q_1 - Y q_2}{(1-Y)c_{v1}\frac{p+\gamma_1\pi_1}{p+\pi_1} + Y c_{v2}\frac{p+\gamma_2\pi_2}{p+\pi_2}}, \\ \mathcal{R}(Y) &= \zeta\,(\rho Y - m_2^{\mathrm{tr}}) + (g_2 - g_1)\big(p, T(p,Y)\big) = 0. \end{aligned} \tag{60}$$

At fixed energy T decreases with Y, since $e_2(p,T) > e_1(p,T)$, and $\partial_T(g_2 - g_1) = -(\mathcal{S}_2 - \mathcal{S}_1) < 0$, so $\mathcal{R}'(Y) = \zeta\rho + (\mathcal{S}_2 - \mathcal{S}_1)|\partial_Y T| > 0$ and Y is unique on its admissible interval $[Y_{\min}, \min(1 - Y_{\min}, ((\rho e)^C/\rho - q_1)/(q_2 - q_1))]$, $Y_{\min} = 10^{-10}$, the upper bound keeping $T > 0$; it is found by Brent's method. If $\mathcal{R}(Y)$ does not change sign on the interval, Y takes the end value: complete condensation or evaporation. Then $\Phi(p) = \rho\sum_k Y_k v_k(p,T) - 1$. For $\zeta = 0$, $g_1 = g_2$, that is $T = T_{\mathrm{sat}}(p)$: the pTg closure below, for $p < p_c$. A state-dependent rate such as (66) is evaluated in each cell from its pT-equilibrium state at $\mathbf{U}^n$ and frozen over the stages, which makes the step first order in time where the transfer is not stiff.

Instantaneous chemical relaxation: pTg closure, $\mu = \theta = \nu = \infty$ $T = T_{\mathrm{sat}}(p)$, Y follows from the energy and $\Phi(p) = \rho\sum_k Y_k v_k - 1$; Y is clamped to $[Y_{\min}, 1 - Y_{\min}]$, with T then from the energy, which coincides with the end-value branch of the finite-rate closure, so Φ is continuous. For $p \geq p_c$ the cell is single-phase, the phase being chosen by comparing $1/\rho$ with $v_1(p_c, T_{\mathrm{sat}}(p_c))$.

Activation and projection Thermal and chemical relaxation are active in interface cells, $\epsilon_I \leq \alpha_1 \leq 1 - \epsilon_I$ with $\epsilon_I = 10^{-4}$, and mass transfer where, in addition, the temperature of the pT-equilibrium state exceeds $T_{\mathrm{sat}}(p)$. Activation is decided once per time step on $\mathbf{U}^n$. An instantaneous relaxation activated inside a stage is an $O(1)$ jump of the state, which the stage combination of the IMEX scheme would extrapolate; for the instantaneous pT and pTg closures $\mathbf{U}^n$ is therefore projected onto their equilibrium manifold by the same closure with $s = 0$, which conserves m_k, or ρ for pTg, $\rho\mathbf{u}$ and E, and a cell whose projection has no admissible solution uses the p closure for that step. The closure type is frozen over the stages.

3.2.4. Solution of the pressure system

With the face velocity and face pressure of Sec. 3.2.1, the fully discrete counterpart of (34) is the system

$$G_{i,j}(P) \equiv \bar{p}^{\mathrm{cl}}_{i,j}\big(s_{i,j}, X^{\mathrm{tr}}_{i,j}, (\rho e)^{C}_{i,j};\ \mu, \theta, \nu\big) - P_{i,j} = 0 \tag{61}$$

for the cell pressures $P_{i,j}$. The nonlinearity of the equations of state and of the relaxation is local to the cell, and the cells are coupled through the face velocities and face pressures, which are affine in the adjacent cell pressures, as in mildly nonlinear systems [18, 33]; the Jacobian is a sparse three-point matrix, five-point in 2D. The system is solved by Newton's method with a finite-difference Jacobian, starting from the pressures of the previous stage. The unknown is the pressure perturbation $P - P_{\mathrm{ref}}$, with P_{ref} the minimum of the initial mixture pressure. The residual of cell (i, j) is scaled by $\mathcal{N}_{i,j} = |P_{i,j}| + K^{W}_{i,j} + P_{E,i,j} + 1$, where K^{W} is the Wood bulk modulus of X^{*}, $\max_k \pi_k$ where X^{*} is not admissible, and $P_E = \sum_k \big(|\varepsilon^{*}_k| + |m^{*}_k q_k|\big) / \sum_k \partial\varepsilon_k/\partial p|_{\alpha_k, m_k}$ is the pressure scale of the energy round-off, with $|m^{*}_k \varepsilon_r| + |\alpha^{*}_k p_r/\Gamma_k|$ in place of $|m^{*}_k q_k|$ for Mie–Grüneisen phases. The natural scale of P is $\hat{P}_{i,j} = \mathcal{N}_{i,j}/J_{i,j}$ with $J_{i,j} = 1 + (K^{W}_{i,j} + P_{E,i,j}) \sum_f (\tau/h_f)\Theta_f$ the scale of $\partial G_{i,j}/\partial P_{i,j}$, Θ_f being the pressure coefficient of u_f of Sec. 3.6, and the finite-difference increments are $10^{-7}\hat{P}$. The linear systems are solved by the Thomas algorithm for non-periodic one-dimensional problems and otherwise by BiCGStab with a Jacobi plus constant-coarse-space preconditioner, converged when the residual norm is below 10^{-13} times that of the right-hand side within 4000 iterations, or below 10^{-10} times after 4000 iterations. The iteration stops when $\max |G_{i,j}|/\mathcal{N}_{i,j} < 10^{-13}$, or when the Newton step satisfies $\max |\Delta P_{i,j}|/(|P_{i,j} - P_{\mathrm{ref}}| + \hat{P}_{i,j}) < 10^{-13}$ and $P + \Delta P$ is taken. Otherwise the step is damped by $\omega = 1, \frac{1}{2}, \ldots, 2^{-39}$ until every local closure is admissible and $\max |G_{i,j}|/\mathcal{N}_{i,j}$ decreases by the factor $1 - 10^{-4}\omega$; if no such ω exists, the iterate is the solution when every $|G_{i,j}|$ lies within $10^{-13}\mathcal{N}_{i,j}$ plus the round-off floor of the closure, and the iteration fails otherwise. At most 60 Newton steps are made.

3.3. Well-balanced property

Proposition 3 (Contacts and phase energies). *(i) Let each phase be a stiffened or Noble–Abel stiffened gas, or a Mie–Grüneisen phase of uniform density. A state with uniform u and p and arbitrary α_k is preserved exactly by the scheme with mechanical relaxation; with thermal or chemical relaxation active this holds if in addition $T_1 = T_2 = T$ is uniform and, with mass transfer, $T \le T_{\mathrm{sat}}(p)$. (ii) If the specific internal energy e_k of a phase is uniform, the explicit step keeps it uniform for any velocity: the phases are compressed at constant specific internal energy, m_k and ε_k being transported by the same linear Rusanov operator with a shared limiter.*

Proof. (i) By (11) $\varepsilon_k = \mathcal{C}_k(p)\alpha_k + (q_k - b_k\mathcal{C}_k(p))m_k$ in every cell, and for a Mie–Grüneisen phase of uniform density m_k and ε_k are proportional to α_k. The shared limiter keeps these linear relations on the faces, and the face velocities equal u, so that $\delta_x[\bar{u}] = 0$, the strain term of (29) vanishes and α_k is advanced by the same linear Rusanov operator (26) as m_k and ε_k. The relations therefore hold for X^{*}, and for the stage combinations of Sec. 3.5, which are linear. With $P \equiv p$ all terms of (39) beyond the averages vanish, $u_f = u$, $p_f = p$, the increments vanish, $s = \delta = 0$, and the closure returns $\alpha_k = \alpha^{*}_k$, $p_k = p$, so that $P \equiv p$ solves (61) and the Newton iteration, started from p, stops at once. With uniform T the densities $\rho_k = 1/v_k(p, T)$ are uniform, X^{*} has $T_k = T$, and the pT and finite-θ closures and the projection return the same state; for $T \le T_{\mathrm{sat}}(p)$ mass transfer is not activated. (ii) The shared limiter keeps $\varepsilon_k = e_k m_k$ on the faces, so that $F^{e}[\varepsilon_k] = e_k F^{e}[m_k]$ and $\varepsilon^{*}_k = e_k m^{*}_k$; only α_k carries the strain term of (29), so that ρ_k changes where $\delta_x[\bar{u}] \neq 0$.

3.4. Asymptotic preserving property

Theorem 4 (Relaxation limits). *Let Δx and τ be fixed and the closure type of each cell be frozen over the stage as in Sec. 3.2.3. Assume that the local closures have non-singular Jacobians at their solutions, that the Jacobian of (61) is non-singular at the solution of the equilibrium closure, and that at this solution no cell lies at a break point of (54) or, with mass transfer, at a bound of Y or at $p \ge p_c$, no face has $u_f = 0$ in (41), and $N_k > 0$ for finite θ. (i) As $\zeta \to 0$ the stage solution converges to that of the corresponding equilibrium closure with an error $O(\zeta)$, and for $\mathbf{U}^n$ on the equilibrium manifold so does $\mathbf{U}^{n+1}$. (ii) For a*

vanishing rate the stage equations reduce to those of the scheme with that relaxation switched off: $\mu = 0$ gives the frozen two-pressure scheme, $\theta = 0$ the p closure and $\nu = 0$ the pT closure.

Proof. (i) Each finite-rate closure of Sec. 3.2.3 depends smoothly on ζ and coincides with the equilibrium closure at $\zeta = 0$, so the implicit function theorem gives an $O(\zeta)$ difference for each cell. Under the assumptions the residual of (61) is smooth in P and ζ near the solution, and its Jacobian with respect to P is non-singular at $\zeta = 0$; the implicit function theorem applied to (61) then gives a discrete solution that differs from the equilibrium one by $O(\zeta)$, with a constant that depends on Δx and τ. Every stage value is the output of a closure and hence, for $\zeta = 0$, an equilibrium state; since the tableau of Sec. 3.5 is stiffly accurate, $\mathbf{U}^{n+1} = \mathbf{U}_2$ is one too. For $\mathbf{U}^n$ on the equilibrium manifold the projection of Sec. 3.2.3 leaves $\mathbf{U}^n$ unchanged, so that both schemes start from the same data and $\mathbf{U}^{n+1}$ differs by $O(\zeta)$. (ii) $\mu = 0$, $\theta = 0$, $\nu = 0$ give $R = 0$, $\tau\mathcal{Q} = 0$, $Y = m_2^{\mathrm{tr}}/\rho$.

The limit schemes are consistent discretisations of the relaxed models. For the p closure, linearising (55) with $\delta = 0$, each phase is isentropic at the common pressure, $\mathrm{d}p = -K_k\,\mathrm{d}\alpha_k/\alpha_k$, and $\sum_k \mathrm{d}\alpha_k = s$ gives $\mathrm{d}\alpha_k = s\,(\alpha_k/K_k)K^W$, $1/K^W = \sum_j \alpha_j/K_j$. With $s = \Delta t\,\partial_x u$ and the transport part of α_k this is

$$\partial_t\alpha_1 + u\,\partial_x\alpha_1 = \alpha_1\Big(\frac{K^W}{K_1} - 1\Big)\partial_x u = \frac{\alpha_1\alpha_2(K_2 - K_1)}{\alpha_1 K_2 + \alpha_2 K_1}\,\partial_x u,$$

the volume-fraction equation of the Kapila model [45, 67]. In the same way the pT and pTg closures discretise the four-equation and homogeneous equilibrium models, whose volume fractions are dependent variables.

Low Mach number limit of the mixture Let u_0, c_0 and ρ_0 be a reference velocity, sound speed and density and $M = u_0/c_0$. With impermeable boundaries and well-prepared data, pressure variations are of the size of the inertial terms, $\rho_0 u_0^2 = M^2\rho_0 c_0^2$, so that $\bar p = \bar p_{(0)} + M^2\bar p_{(2)} + \dots$ with $\bar p_{(2)} = O(\rho_0 c_0^2)$. In the momentum equation the gradient of $\bar p_{(0)}$ would exceed the inertia by M^{-2}, hence $\partial_x\bar p_{(0)} = 0$; integrating the pressure equation of the mixture, $\partial_t\bar p + u\,\partial_x\bar p + K^W\partial_x u = 0$, over the domain then gives $\bar p_{(0)} = \text{const}$ and $\partial_x u_{(0)} = 0$: the incompressible limit of the mixture.

Proposition 5 (Formal low Mach number limit). *For fixed Δx, Δt given by (32) and data well prepared at both time levels, $P = P_0 + M^2P_2$, formally as $M \to 0$: the face velocities are discretely divergence-free to $O(M^2)$, P_2 solves the discrete Poisson equation $\delta_x\big[\bar u_f^* - \tau\,\Delta_f P_2/(\bar\rho_f\Delta x)\big] = O(M^2)$, with the analogous y-term in two dimensions, the Laplace terms of (39) vanish once $M < a_{ii}C$, and the limit scheme is explicit convection plus a projection, with a time step independent of M.*

Proof. Scale pressures by $\rho_0 c_0^2$, so that the pressure coefficient of the face velocity, $\tau/(\bar\rho_f\Delta x)$ in dimensional form, carries a factor M^{-2} relative to the inertia. Linearising the closure about X^{tr}, the pressure equation (61) reads, in every cell,

$$P - P^{\mathrm{tr}} = -K^W s + \dots, \qquad s = \frac{\tau}{\Delta x}\,\delta_x[u_f], \qquad u_f = \bar u_f^* - \frac{\tau}{M^2\bar\rho_f}\,\frac{\Delta_f P}{\Delta x} + \dots,$$

with $K^W = O(1)$ and P^{tr} the mixture pressure of X^{tr}, which is to first order P^n advected with the flow, as in (35), so that $P - P^{\mathrm{tr}} = O(M^2)$ for well-prepared data. Insert $P = P_0 + M^2P_2$ with P_0 constant. The term $K^W s$ must be $O(M^2)$, so $\delta_x[u_f] = O(M^2)$: the face velocities are discretely divergence-free. The pressure gradient in u_f is $\Delta_f P/(M^2\Delta x) = \Delta_f P_2/\Delta x$, and the order-one balance gives the compact Poisson equation of the statement, i.e. a projection of the explicit face velocities. In the momentum update p_f is a weighted average of P, P_0 cancels in $\delta_x[p_f]$, and the cell velocity is corrected by $-\tau\,\delta_x[p_{f,2}]/(\rho\Delta x)$, with $p_{f,2}$ the weighted face average of P_2. The acoustic Courant number is $c\tau/\Delta x = a_{ii}C\,c/\max|u| = O(1/M)$, so $\varphi = (1 - c\tau/\Delta x)^+ = 0$ once $M < a_{ii}C$. The explicit Rusanov dissipation uses the material speed and the closures return states on the equilibrium manifold, both independently of M. The limit scheme is therefore explicit convection followed by a projection of the face velocities, with a time step set by (32).

3.5. Second-order time discretisation

The scheme of Secs. 3.1 and 3.2 is first order in time. Second order is obtained with the semi-implicit IMEX Runge–Kutta schemes of Boscarino, Filbet and Russo [8], for which the time linearisation introduced by the splitting, explicit velocity in the transport fluxes and implicit velocity increment, gives the semi-implicit structure directly. The semi-discrete system is written in autonomous form as

$$\partial_t \mathbf{U} = \mathcal{H}\big(\mathbf{U}_E(t), \mathbf{U}_I(t)\big), \tag{62}$$

where $\mathcal{H}$ is the spatial discretisation of the split system, $\mathbf{U}_E$ the state on which the explicit transport is evaluated and $\mathbf{U}_I$ the state of the pressure–relaxation sub-system. An IMEX Runge–Kutta scheme [3, 64] is defined by a pair of Butcher tableaux $(\tilde{A}, \tilde{\mathbf{b}}, \tilde{\mathbf{c}})$ and $(A, \mathbf{b}, \mathbf{c})$, $\tilde{A}$ strictly lower triangular and A lower triangular. We use the stiffly accurate, L-stable LSDIRK2(2,2,2) scheme [4, 8],

$$\begin{array}{c|cc} 0 & 0 & 0 \\ \hat{c} & \hat{c} & 0 \\ \hline & 1-\beta & \beta \end{array} \qquad \begin{array}{c|cc} \beta & \beta & 0 \\ 1 & 1-\beta & \beta \\ \hline & 1-\beta & \beta \end{array}, \qquad \beta = 1 - \tfrac{1}{\sqrt{2}}, \quad \hat{c} = \tfrac{1}{2\beta}. \tag{63}$$

With $\mathbf{U}_E^n = \mathbf{U}_I^n = \mathbf{U}^n$, the stages $i = 1, 2$ are

$$\begin{aligned} \mathbf{U}_{E,i} &= \mathbf{U}^n + \Delta t \sum_{j<i} \tilde{a}_{ij} \mathcal{K}_j, \qquad \tilde{\mathbf{U}}_{I,i} = \mathbf{U}^n + \Delta t \sum_{j<i} a_{ij} \mathcal{K}_j, \\ \mathcal{K}_i &= \mathcal{H}\big(\mathbf{U}_{E,i},\ \tilde{\mathbf{U}}_{I,i} + \Delta t\, a_{ii} \mathcal{K}_i\big). \end{aligned} \tag{64}$$

Each stage is one application of the first-order scheme: the explicit fluxes (25) are evaluated on $\mathbf{U}_{E,i}$, the update starts from $\tilde{\mathbf{U}}_{I,i}$ instead of $\mathbf{U}^n$, and the step, explicit transport included, has the length $\tau = a_{ii}\Delta t$, giving the stage value $\mathbf{U}_i$ and $\mathcal{K}_i = (\mathbf{U}_i - \tilde{\mathbf{U}}_{I,i})/\tau$. Since $\mathbf{b}$ is the last row of A, $\mathbf{U}^{n+1} = \mathbf{U}_2$. The first step is the smaller of (32) and the acoustic step $0.4\big(\max_\Omega(|u| + c_f)/\Delta x + \max_\Omega(|v| + c_f)/\Delta y\big)^{-1}$, as in [7], and no step may exceed 1.2 times the previous one, so that (32) is reached after about $\ln(0.5/M_\Omega)/\ln 1.2$ steps, $M_\Omega = \max_\Omega |u| / \max_\Omega(|u| + c_f)$; the steps limited by this rule, up to the first step that it does not limit, use one backward-Euler stage. If in some stage the mixture density of $\mathbf{U}_{E,i}$ or X^* is non-positive in a cell, a local closure has no admissible solution or the Newton iteration fails, the step is repeated with half the time step and the same tableau. With a prescribed fixed time step the start-up rule is not applied.

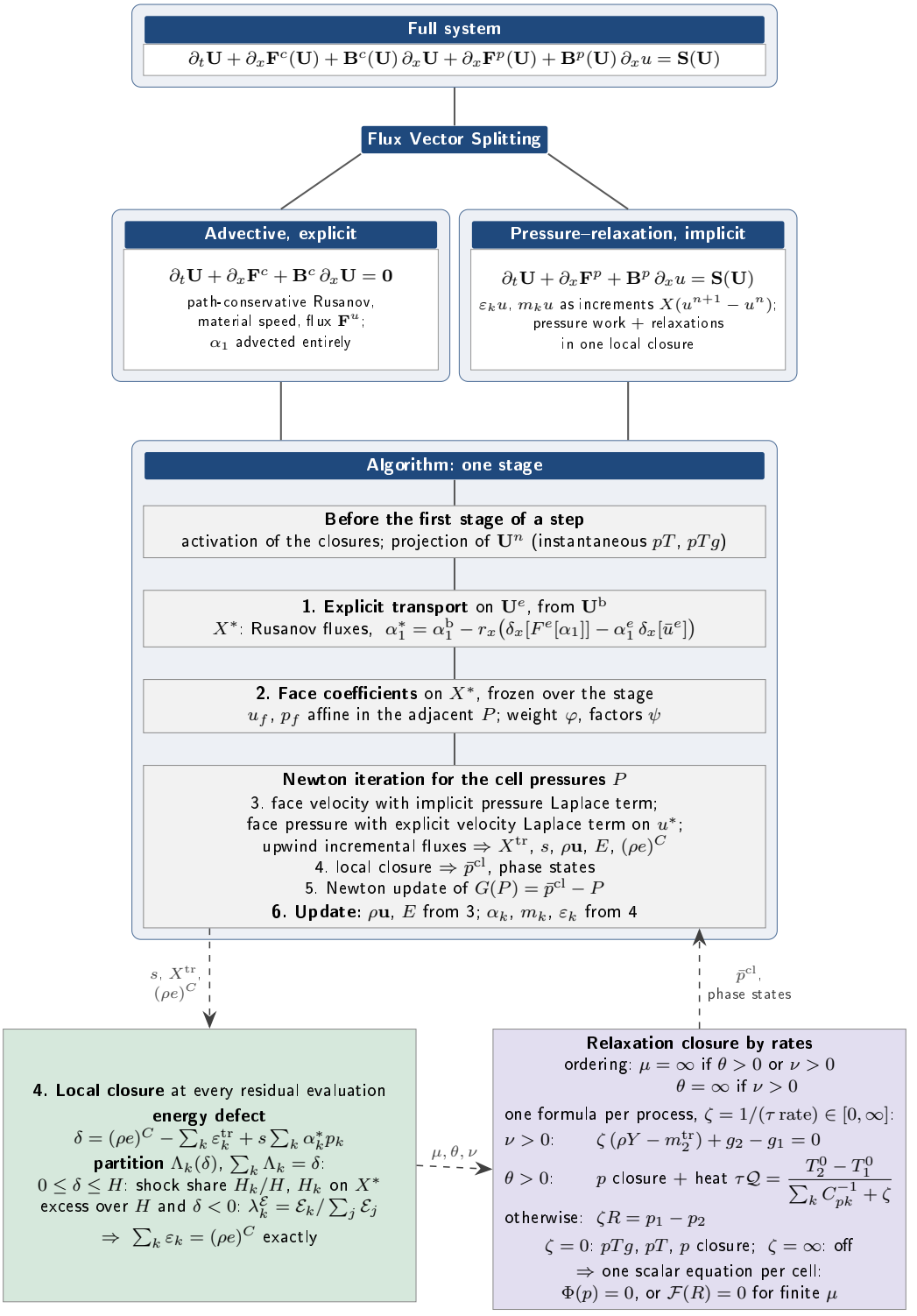


Figure 1: Schematic of the semi-implicit scheme. The Toro–Vázquez-Cendón splitting separates the advective sub-system, advanced explicitly with the old-velocity part of the internal energy fluxes, from the pressure–relaxation sub-system, solved implicitly with the increments of the mass and internal energy fluxes. Every residual evaluation of the pressure system calls the local closure of each cell, which distributes the energy defect δ between the phases, and the relaxation closure selected by the rates.

3.6. Fully discrete scheme in two dimensions

One stage of the scheme, with the explicit state $\mathbf{U}^e = \mathbf{U}_{E,i}$, the update base $\mathbf{U}^{\mathrm{b}} = \tilde{\mathbf{U}}_{I,i}$ and the implicit step $\tau = a_{ii}\Delta t$, or $\mathbf{U}^e = \mathbf{U}^{\mathrm{b}} = \mathbf{U}^n$ and $\tau = \Delta t$ in a first-order step, consists of the following steps, with $r_x = \tau/\Delta x$ and $r_y = \tau/\Delta y$. Before the first stage of a step the closures are activated and $\mathbf{U}^n$ is projected as in Sec. 3.2.3.

1. *Explicit transport.* The face values are reconstructed on $\mathbf{U}^e$ by (30) with the cap (31), and every transported quantity is advanced with the Rusanov fluxes (26),

$$X^*_{i,j} = X^{\mathrm{b}}_{i,j} - r_x\,\delta_x\big[F^e[X]\big]_{i,j} - r_y\,\delta_y\big[G^e[X]\big]_{i,j}, \quad X \in \{m_k, \varepsilon_k, \rho u, \rho v, E\},$$

and the volume fraction by the path-conservative update (29) with base α_1^{b} and the cell's own volume fraction α_1^e of $\mathbf{U}^e$ in the strain term,

$$\begin{aligned}\alpha^*_{1,i,j} = \alpha^{\mathrm{b}}_{1,i,j} &- r_x\Big(\delta_x\big[F^e[\alpha_1]\big]_{i,j} - \alpha^e_{1,i,j}\,\delta_x[\bar{u}^e]_{i,j}\Big)\\ &- r_y\Big(\delta_y\big[G^e[\alpha_1]\big]_{i,j} - \alpha^e_{1,i,j}\,\delta_y[\bar{v}^e]_{i,j}\Big), \qquad \alpha^*_2 = 1-\alpha^*_1.\end{aligned}$$

2. *Face coefficients.* On every face the quantities of Sec. 3.2.1 are computed from X^*, the limiter factors of the reconstruction of $\mathbf{U}^e$ and the pressures of the previous stage: the averages $\bar{u}^*_f$, $\bar{\rho}_f$, the impedances Z, the weight (42), the factors (43) and the velocity jump (40). They are frozen during the solution of the stage, so that the face velocity and face pressure (39) are affine in the two adjacent cell pressures,

$$u_f = \bar{u}^*_f - \Theta_f\,(P_R - P_L), \qquad p_f = W_f P_L + (1-W_f)P_R - \varpi_f,$$

with $\Theta_f = \tau/(\bar{\rho}_f h) + \eta^P_f/h$, $W_f = \rho^*_R/(\rho^*_L + \rho^*_R)$ and $\varpi_f = \eta^u_f D^*_f/h$, $h = \Delta x$ or Δy. At a non-periodic boundary the ghost cells take the pressure of the adjacent cell, or the given pressure at the outlet of Sec. 4.5, and their state is a copy of the boundary cells at transmissive boundaries and their mirror image at walls, with the normal velocity $2u_w - u$ for a wall velocity u_w; on a wall face $u_f = u_w$. At periodic boundaries the ghost cells and their pressures are those of the opposite side.

3. *Incremental fluxes and conservative updates.* For given cell pressures, the incremental fluxes (41) are split into their donor parts $I_f = I^L_f + I^R_f$; the incremental mass fluxes are $\hat{M}^{L,R}_f = \sum_k I^{L,R}_f[m_k]$, $\hat{M}_f = \hat{M}^L_f + \hat{M}^R_f$. Then

$$\begin{aligned}&X^{\mathrm{tr}}_{i,j} = X^*_{i,j} - r_x\,\delta_x\big[I[X]\big]_{i,j} - r_y\,\delta_y\big[I[X]\big]_{i,j}, \qquad X \in \{m_k, \varepsilon_k\},\\ &s_{i,j} = r_x\,\delta_x[u_f]_{i,j} + r_y\,\delta_y[v_f]_{i,j}, \qquad \alpha^{\mathrm{tr}}_{k,i,j} = \alpha^*_{k,i,j}(1-s_{i,j}),\\ &\rho^{\mathrm{new}}_{i,j} = \textstyle\sum_k m^{\mathrm{tr}}_{k,i,j}.\end{aligned}$$

Let $u_\perp$ be the normal face velocity, u_f on x-faces and v_f on y-faces, and $u_{\parallel,L}$, $u_{\parallel,R}$ the tangential velocities of X^* in the two cells of the face, v^* on x-faces and u^* on y-faces. The momentum is advanced with the face pressure and the incremental momentum flux, normal and tangential,

$$\begin{aligned}(\rho u)^{\mathrm{new}}_{i,j} &= (\rho u)^*_{i,j} - r_x\,\delta_x\big[p_f + u_\perp \hat{M}_f\big]_{i,j} - r_y\,\delta_y\big[u_{\parallel,L}\hat{M}^L_f + u_{\parallel,R}\hat{M}^R_f\big]_{i,j},\\ (\rho v)^{\mathrm{new}}_{i,j} &= (\rho v)^*_{i,j} - r_x\,\delta_x\big[u_{\parallel,L}\hat{M}^L_f + u_{\parallel,R}\hat{M}^R_f\big]_{i,j} - r_y\,\delta_y\big[p_f + u_\perp \hat{M}_f\big]_{i,j},\end{aligned}$$

and the total energy with the energy flux

$$\begin{aligned}F^I_f[E] &= \textstyle\sum_k I_f[\varepsilon_k] + \frac{1}{2}u^2_\perp \hat{M}_f + \frac{1}{2}\big(u^2_{\parallel,L}\hat{M}^L_f + u^2_{\parallel,R}\hat{M}^R_f\big) + p_f u_\perp,\\ E^{\mathrm{new}}_{i,j} &= E^*_{i,j} - r_x\,\delta_x\big[F^I[E]\big]_{i,j} - r_y\,\delta_y\big[F^I[E]\big]_{i,j}.\end{aligned}$$

In each cell the conservative mixture internal energy is $(\rho e)^C = E^{\mathrm{new}} - |(\rho\mathbf{u})^{\mathrm{new}}|^2/(2\rho^{\mathrm{new}})$.

4. *Local closure.* In every cell the defect (45) is partitioned by (54), with the physical budget (51) computed on X^* after step 1 and frozen during the stage, and the relaxation closure of Sec. 3.2.3 returns $\bar{p}^{\mathrm{cl}}_{i,j}$ and the phase states.

5. *Pressure system.* The system (61) is solved for the cell pressures by the Newton method of Sec. 3.2.4; every residual evaluation repeats steps 3 and 4.
6. *Update.* At the converged pressures, the stage value consists of the momentum and the total energy of step 3 and the volume fractions, partial densities and phase energies of the closure of step 4.

Steps 1–6 give the stage value $\mathbf{U}_i$ of (64), step 1 being its explicit part and steps 2–6 its implicit part.

4. Numerical results

In this section we validate the proposed semi-implicit scheme on a suite of test problems, following the order in which the ingredients of the method are introduced. We begin by assessing the accuracy of the full scheme on a smooth two-phase vortex, measuring the experimental order of convergence as the Mach number is decreased. We then turn to one-dimensional problems: the advection of material interfaces, which checks the well-balanced property; single-phase Riemann problems over a wide range of Mach numbers; shock tubes with mechanical relaxation, with stiffened-gas and Mie–Grüneisen equations of state and with a shock in a two-phase mixture, which checks the distribution of the energy defect; and expansion, evaporation and depressurisation problems with thermal and chemical relaxation, at instantaneous and finite rates. The low Mach number limit of the mixture is then examined on the Taylor–Green vortex. Finally, we consider two-dimensional problems with interfaces, shocks, cavitation and phase transition.

Unless otherwise stated, all computations use the scheme of Sec. 3 with LSDIRK2, the material CFL condition (32) with $C = 0.2$, minmod reconstruction, a local Rusanov speed on each face and the Newton tolerance of Sec. 3.2.4 and the start-up rule of Sec. 3.5. Thermal and chemical relaxation are active in cells with $10^{-4} \le \alpha_1 \le 1 - 10^{-4}$. Where a phase is absent its volume fraction is set to 10^{-8}, 10^{-7} or 10^{-6} as indicated; initial data in two dimensions are cell averages computed with 4×4 sub-cells. Units are SI unless stated otherwise.

Mechanical relaxation is instantaneous in all tests; finite rates are used for thermal and chemical relaxation only. The relaxation levels are denoted as in Pelanti and Shyue [68]. p: instantaneous mechanical relaxation only. p–pT: in addition, instantaneous thermal relaxation in interface cells. p–pT–pTg: in addition, instantaneous mass transfer in interface cells where the pT-equilibrium temperature exceeds $T_{\mathrm{sat}}(p)$. A finite rate replaces the corresponding instantaneous relaxation and is marked (f). Reference solutions are exact Riemann solutions of the relaxed limit models: the five-equation model for p, the four-equation model for p–pT, and the $pT \to pTg$ wave fan and the homogeneous equilibrium model for instantaneous mass transfer.

4.1. Accuracy of the numerical scheme: two-phase isentropic vortex

This test of Battisti and Boscheri [7, Sec. 4.1] measures the order of accuracy for a smooth two-phase flow at decreasing Mach number. The domain is $[-5, 5]^2$ with periodic boundaries; phase 1 is an ideal gas with $\gamma_1 = 1.35$ and phase 2 an ideal gas with $\gamma_2 = 1.4$. With $r^2 = x^2 + y^2$, $\varepsilon = 5$ and $\gamma = 1.4$,

$$
\begin{aligned}
&u = 2 - \frac{\varepsilon}{2\pi} e^{(1-r^2)/2}\, y, \qquad v = 2 + \frac{\varepsilon}{2\pi} e^{(1-r^2)/2}\, x, \qquad T = p_0 - \frac{(\gamma-1)\varepsilon^2}{8\gamma\pi^2} e^{1-r^2}, \\
&\rho_1 = \rho_2 = \left(\frac{T}{p_0}\right)^{1/(\gamma-1)}, \qquad p_1 = p_2 = \rho_1 T, \qquad \alpha_2 = \frac{1}{2} + \frac{1}{4} e^{(1-r^2)/2}.
\end{aligned}
\tag{65}
$$

The phases have equal pressure and density, the velocity field is divergence-free and α_2 and the entropy are constant along streamlines, so the exact solution is the translation of (65) with velocity $(2, 2)$; at $t = 5$ it coincides with the initial data. The background pressure is $p_0 = 1, 10, 10^2, 10^3, 10^4$, which gives $M = 2\sqrt{2}/\sqrt{\gamma p_0}$ from 2.4 to 0.024, and the grids have N^2 cells with $N = 32, 64, 128, 256$. Two initial data are used. In the first the cell averages of (65) are taken. In the second, well-prepared, the velocity is the central difference of the stream function $\psi = \frac{\varepsilon}{2\pi} e^{(1-r^2)/2}$ at the cell centres, $u_{i,j} = 2 + (\psi_{i,j+1} - \psi_{i,j-1})/(2\Delta y)$, $v_{i,j} = 2 - (\psi_{i+1,j} - \psi_{i-1,j})/(2\Delta x)$, which is discretely divergence-free; the other variables are cell averages. In both cases the errors are measured in L_2 against the 4×4-point cell averages of the exact solution. The slopes are the unlimited central differences.

Table 1
Single-phase Riemann problems [80]: left and right states and final time.

problem	fluid	$(\rho, u, p)_L$	$(\rho, u, p)_R$	t_{end}
low-Mach shock tube	water	$(1000,\ 0,\ 10^8)$	$(1000,\ 15,\ 0.98 \times 10^8)$	10^{-4}
liquid–liquid shock tube	water	$(1000,\ 0,\ 10^6)$	$(1000,\ 0,\ 10^5)$	1.5×10^{-4}
Lax shock tube	air	$(0.445,\ 0.698,\ 3.528)$	$(0.5,\ 0,\ 0.571)$	0.14
strong shock tube	air	$(1,\ 0,\ 10^{10})$	$(0.125,\ 0,\ 0.1)$	2.5×10^{-6}
two rarefactions	air	$(1,\ -1,\ 0.4)$	$(1,\ 1,\ 0.4)$	0.15

4.2. One-dimensional Riemann problems

The following problems in one space dimension test the scheme on material interfaces, on single-fluid waves over a wide range of Mach numbers and on shocks with mechanical relaxation.

4.2.1. Interface advection

A water–air interface is advected in $[0, 1]$ with transmissive boundaries [78, Sec. 4.1]. Water is a stiffened gas with $\gamma = 4.4$, $\pi = 6 \times 10^8$ Pa, air an ideal gas with $\gamma = 1.4$. Initially $u = 100\,\text{m/s}$, $p = 10^5$ Pa, $\rho_{\text{water}} = 1000$, $\rho_{\text{air}} = 10$, and $\alpha_{\text{water}} = 1 - 10^{-8}$ for $x < 0.5$, 10^{-8} otherwise. The computation uses 200 cells and ends at $t = 2.79\,\text{ms}$, when the interface is at $x = 0.779$. Two low-Mach variants follow Schropff et al. [80]: the same fluids with $u = 1\,\text{m/s}$, $\rho_{\text{air}} = 0.01$, $\alpha_{\text{water}} = 1 - 10^{-7}$ for $x < 0.25$ and 10^{-7} otherwise, $t_{\text{end}} = 0.25\,\text{s}$, so that $M = |u|/c$ lies between 2.7×10^{-4} in the air and 6×10^{-4} in the water; and a single gas, air, with a density step $\rho = 1000$ for $x < 0.25$ and $\rho = 0.01$ otherwise, $u = 1\,\text{m/s}$, $p = 10^5$ Pa, $t_{\text{end}} = 0.5\,\text{s}$, so that M ranges from 2.7×10^{-4} to 8.5×10^{-2}. Both use 200 and 1000 cells. In all three cases uniform velocity and pressure must be preserved (Proposition 3).

4.2.2. Single-phase Riemann problems

The Riemann problems of Table 1 [80, Secs. 5.1, 5.1.1] test the scheme for a single fluid over a wide range of Mach numbers. A single fluid is represented by two identical phases in the same state, with $\alpha_1 = 1 - 10^{-8}$. Air is an ideal gas with $\gamma = 1.4$ and water the stiffened gas of Sec. 4.2.1. The domain is $[0, 1]$ with transmissive boundaries, the initial discontinuity is at $x = 0.5$, and 200 and 1000 cells are used. The reference is the exact solution of the Euler equations.

4.2.3. Shock tubes with mechanical relaxation

The following Riemann problems on $[0, 1]$, or $[-10, 10]$ for the detonation products, with transmissive boundaries are computed with the p relaxation and compared with the exact solution of the five-equation model. The fluids are at rest initially.

Water–air shock tube Water and air as in Sec. 4.2.1 [78, Sec. 4.3.1]. For $x < 0.75$, water with $\alpha_{\text{water}} = 1 - 10^{-6}$, $\rho = 1000$, $p = 10^9$ Pa; for $x > 0.75$, air with $\alpha_{\text{air}} = 1 - 10^{-6}$, $\rho = 1$, $p = 10^5$ Pa; $t_{\text{end}} = 240\,\mu\text{s}$, with 1000 and 100 cells.

Nitromethane shock tube Both phases are liquid nitromethane with the Cochran–Chan law (13) [78, Sec. 4.2] with $\Gamma = 1.19$, $\rho_0 = 1134$, $A_1 = 0.819181\,\text{GPa}$, $A_2 = 1.50835\,\text{GPa}$, $E_1 = 4.52969$, $E_2 = 1.42144$. Phase 1 has $\rho_1 = 1134$ and phase 2 $\rho_2 = 1200$; $\alpha_1 = 1 - 10^{-8}$, $p = 2 \times 10^{10}$ Pa for $x < 0.5$ and $\alpha_1 = 10^{-8}$, $p = 2 \times 10^5$ Pa otherwise. 500 cells, $t_{\text{end}} = 67\,\mu\text{s}$. The nonlinear reference curve makes the contact discontinuity a test of pressure and velocity oscillations.

Detonation products–water shock tube Detonation products follow the JWL law (13) [67, Sec. 8.1.1] with $\Gamma = 0.25$, $\rho_0 = 1840$, $A = 854.5\,\text{GPa}$, $B = 20.5\,\text{GPa}$, $R_1 = 4.6$, $R_2 = 1.35$; water is NASG with $(\gamma, \pi, q, q', c_v, b) = (1.187,\ 7.028 \times 10^8,\ -1177788,\ 0,\ 3610,\ 6.61 \times 10^{-4})$. On $[-10, 10]$, the products $(\alpha = 1 - 10^{-8},\ \rho = 2000,\ p = 4.6406 \times 10^{10}\,\text{Pa})$ fill $x < 0$ and water $(\rho = 1044,\ p = 10^5\,\text{Pa})$ $x > 0$. 500 cells, $t_{\text{end}} = 0.9\,\text{ms}$.

Epoxy–spinel mixture A mixture of epoxy (SG, $\gamma = 2.43$, $\pi = 5.3 \times 10^9\,\mathrm{Pa}$) and spinel (SG, $\gamma = 1.62$, $\pi = 1.41 \times 10^{11}\,\mathrm{Pa}$) with $\alpha_{\mathrm{epoxy}} = 0.5954$, $\rho_{\mathrm{epoxy}} = 1185$, $\rho_{\mathrm{spinel}} = 3622$ everywhere [78, Sec. 5.2]. The pressure is $2 \times 10^{11}\,\mathrm{Pa}$ for $x < 0.6$ and $10^5\,\mathrm{Pa}$ otherwise; 500 cells, $t_{\mathrm{end}} = 34\,\mu\mathrm{s}$, global Rusanov speed. Both phases are compressed through the same shock; the test checks the partition of the energy defect of Sec. 3.2.2.

4.3. Water cavitation tube

Two streams of liquid water moving apart from $x = 0.5$ in $[0, 1]$, with transmissive boundaries, produce a strong expansion at the centre [67, Sec. 8.2.1]. Water is NASG with liquid (1.187, 7.028×10^8, -1177788, 0, 3610, 6.61×10^{-4}) and vapour (1.467, 0, 2077616, 14317, 955, 0). Initially $\alpha_v = 10^{-2}$, both phases at $p = 10^5\,\mathrm{Pa}$ and $T_0 = 353\,\mathrm{K}$ with $\rho_k = \rho_k(p, T_0)$, and $u = -2\,\mathrm{m/s}$ for $x < 0.5$, $+2\,\mathrm{m/s}$ otherwise; 5000 cells, $t_{\mathrm{end}} = 3\,\mathrm{ms}$. The same configuration is computed with five levels: p; p–$pT(f)$ with $\theta = 2000$; p–pT; p–pT–$pTg(f)$ with $\nu = 10^{-4}$; p–pT–pTg.

4.4. Dodecane liquid–vapour shock tube

Liquid dodecane at high pressure expands into its vapour [67, Sec. 8.2.2]. Dodecane is NASG with liquid (1.206, 1.681×10^8, -996054, 0, 2532, 7.51×10^{-4}) and vapour (1.021, 0, -384592, -4301, 2274, 0). For $x < 0.75$, liquid with $\alpha_l = 1 - 10^{-8}$, $\rho_l = 500$, $p = 10^8\,\mathrm{Pa}$; for $x > 0.75$, vapour with $\alpha_v = 1 - 10^{-8}$, $\rho_v = 2$, $p = 10^5\,\mathrm{Pa}$; transmissive boundaries, 2000 cells, $t_{\mathrm{end}} = 473\,\mu\mathrm{s}$; levels p and p–pT–pTg.

4.5. Barták rapid depressurisation

A pipe of length $1.7\,\mathrm{m}$ filled with hot pressurised water is opened at one end [67, Sec. 8.2.4], after the experiment of Barták [6]. Water is NASG with liquid (1.387, 8.899×10^8, -1244191, 0, 3202, 4.78×10^{-4}) and vapour (1.954, 0, 2287484, 6417, 462, 0). Initially $p = 12.5\,\mathrm{MPa}$, $T = 563.15\,\mathrm{K}$, $u = 0$, $\alpha_v = 10^{-3}$ and $\rho_k = \rho_k(p, T)$. The right end $x = 1.7$ is a wall; at the left end the pressure $10^5\,\mathrm{Pa}$ is imposed: the ghost cells keep α, m_k and u of the adjacent cell with both phase pressures equal to $10^5\,\mathrm{Pa}$, and this pressure is a given value in the pressure system. 1000 cells, $t_{\mathrm{end}} = 17.5\,\mathrm{ms}$; pressure and vapour mass fraction are recorded at $x = 48\,\mathrm{mm}$. Three cases are computed with instantaneous thermal relaxation: no mass transfer (p–pT), instantaneous mass transfer (p–pT–pTg), and the finite rate

$$\nu = C_r\,\alpha_v^{0.6}\Big(\frac{p_{\mathrm{sat}}(T) - p}{p_{\mathrm{crit}} - p_{\mathrm{sat}}(T)}\Big)^{1.76} \quad (p < p_{\mathrm{sat}}(T)), \qquad \nu = 0 \text{ otherwise}, \tag{66}$$

with $C_r = 1$ and $p_{\mathrm{crit}} = 22.064\,\mathrm{MPa}$, evaluated in each cell from its pT-equilibrium state at the beginning of the step and frozen over the stages. With instantaneous mass transfer the solution at the probe is a simple wave of the homogeneous equilibrium model, which is computed exactly.

4.6. Low Mach number limit: Taylor–Green vortex

This test of Battisti and Boscheri [7, Sec. 4.4] checks the low-Mach limit of Sec. 3.4. On $[0, 2\pi]^2$ with periodic boundaries both phases are ideal gases with $\gamma = 1.4$, and

$$\rho_1 = \rho_2 = 1, \qquad u = \sin x \cos y, \qquad v = -\cos x \sin y, \qquad p_1 = p_2 = p_0 + \tfrac{1}{4}(\cos 2x + \cos 2y), \tag{67}$$

a stationary solution of the incompressible Euler equations [83]. The volume fraction is $\alpha_2 = \frac{1}{2}$ in the single-phase case and $\alpha_2 = \frac{1}{2} + \frac{1}{10}(\cos 2x + \cos 2y)$ in the two-phase case. The background pressure $p_0 = 10^2, 10^4, \ldots, 10^{12}$ gives $M = 1/\sqrt{1.4\,p_0}$ from 8.45×10^{-2} to 8.45×10^{-7}. On a 50×50 grid the solution is computed to $t = 0.2$ with the fixed time step $\Delta t = 2.5 \times 10^{-3}$, independent of M; the order of accuracy is measured for $p_0 = 10^6$ and 10^{10} on $N = 32, \ldots, 256$ with $\Delta t = 0.08/N$. Both computations are made with unlimited central slopes and with minmod. The errors are measured in L_2 against the 4×4-point cell averages of (67), together with the discrete divergence of the face velocities.

4.7. Shock–helium bubble interaction

A Mach 1.5 shock in air hits a helium bubble [78, Sec. 4.6], after the experiment of Layes and Le Métayer [49]. The domain is $[0, 0.3] \times [0, 0.08]$ with walls at $y = 0$ and $y = 0.08$ and transmissive boundaries in x;

air is an ideal gas with $\gamma = 1.4$, helium with $\gamma = 1.67$. The bubble has diameter 0.04 and centre $(0.2, 0.04)$; inside $\alpha_{\text{air}} = 10^{-6}$, outside $1 - 10^{-6}$. Ahead of the shock, air has $\rho = 1.29$, $p = 101325\,\text{Pa}$ and helium $\rho = 0.167$, both at rest. The shock is at $x = 0.221$ and moves to the left; behind it the air state follows from the Rankine–Hugoniot relations of the ideal gas at Mach 1.5. 600×160 cells, $t_{\text{end}} = 600\,\mu\text{s}$, p relaxation.

4.8. Cavitating Richtmyer–Meshkov instability

A water column with a curved surface moving at $-200\,\text{m/s}$ hits a wall [78, Sec. 4.7]. On $[0,3] \times [0,1]$, water (SG, $\gamma = 4.4$, $\pi = 6 \times 10^8\,\text{Pa}$, $\rho = 1000$) occupies $x < 1.2 + \sqrt{0.36 - (y - 0.5)^2}$ and gas (ideal, $\gamma = 1.8$, $\rho = 100$) the rest, with $\alpha = 1 - 10^{-6}$ for the present phase; everywhere $u = -200\,\text{m/s}$ and $p = 101325\,\text{Pa}$. The left, bottom and top boundaries are walls and the right boundary is transmissive. 300×100 cells, the grid of Pelanti and Shyue [68, Sec. 5.3], $t_{\text{end}} = 8.6\,\text{ms}$, p relaxation, global Rusanov speed.

4.9. Compression of a vapour bubble

A dodecane vapour bubble in liquid dodecane is compressed by a piston [68, Sec. 5.4]. Dodecane is SG with liquid (2.35, 4×10^8, -775.269×10^3, 0, 1077.7) and vapour (1.025, 0, -237.547×10^3, -24.4×10^3, 1956.45). On $[0,1]^2$ a bubble of radius 0.2 is centred at $(0.5, 0.5)$; initially $p = 10^5\,\text{Pa}$ and $T = 600\,\text{K}$, so $\rho_l = 458.338$ and $\rho_v = 3.408$, with $\alpha_l = 10^{-6}$ inside the bubble and $\alpha_v = 10^{-6}$ outside, at rest. All boundaries are walls; the left wall moves with velocity $100\,\text{m/s}$ in $+x$; the grid is fixed and the wall boundary condition imposes this normal velocity. 200×200 cells, $t_{\text{end}} = 1.2\,\text{ms}$, p–pT–pTg relaxation, global Rusanov speed.

4.10. Underwater explosion near a wall

A bubble of detonation products expands in water below a rigid wall [67, Sec. 8.1.2]. The states and the equations of state are those of the detonation products–water shock tube of Sec. 4.2.3: products with $\rho = 2000$, $p = 4.6406 \times 10^{10}\,\text{Pa}$ and $\alpha = 1 - 10^{-6}$ inside a circle of radius 0.05 centred at $(0, -0.32)$, water with $\rho = 1044$, $p = 10^5\,\text{Pa}$ outside. The domain is $[-1.35, 1.35] \times [-1.5, 0]$ with a wall at $y = 0$ and transmissive boundaries elsewhere. 361×200 cells, half the resolution of [67], $t_{\text{end}} = 0.4\,\text{ms}$, p relaxation; the pressure is recorded at the wall centre $(0, 0)$.

5. Conclusions

We have presented a second-order semi-implicit IMEX finite volume scheme for the single-velocity six-equation two-phase model with total energy, for stiffened-gas and Noble–Abel stiffened-gas phases with mechanical, thermal and chemical relaxation and for Mie–Grüneisen phases with instantaneous mechanical relaxation, in one and two space dimensions.

The two stiffnesses of the model, the acoustic waves at low Mach numbers and the relaxation processes up to infinite rates, are treated in a single implicit step. The implicit pressure work and the three relaxation processes, each at a finite or infinite rate, a slower process being active only when the faster ones are instantaneous [67], enter one local closure per cell, which reduces to a scalar equation in the cell pressure or, for finite mechanical relaxation, in the volume increment; each stage then reduces to a nonlinear system for the cell pressures with a three-point Jacobian, five-point in 2D, solved by Newton's method. The time step is restricted by the flow velocity only, whatever the Mach number and the relaxation rates. We have proved that, at fixed mesh and under non-degeneracy assumptions, the scheme converges, for data on the equilibrium manifold, to the corresponding equilibrium scheme as each relaxation rate tends to infinity and reduces to the scheme without that relaxation as the rate vanishes; formally, it is asymptotic preserving in the low Mach number limit of the mixture.

At shocks, the difference between the conservative and the non-conservative mixture internal energy is distributed between the phases according to the shock relations of the model: the physical part with the constant share for which the shock profile of the relaxed model ends on the Hugoniot of each phase, computed by shooting, which for weak shocks reduces to the unique consistent local share, and the discretisation error with the unique linear share that keeps both phases admissible whenever possible. No calibrated heat exchange is needed.

The flux is split following Toro and Vázquez-Cendón, with the mass and internal energy fluxes evaluated at the new velocity: all quantities are transported explicitly with the old velocity by the path-conservative

Rusanov scheme, whose dissipation speed is the material speed, the volume fraction being advected entirely in this step, while the pressure, its work on each phase and the increments of the mass and internal energy fluxes from the old to the new velocity are treated implicitly. The central implicit operator is supplemented by Laplace dissipation of the pressure and the velocity with acoustic coefficients, weighted by the deficit of the backward-Euler viscosity with respect to the upwind one, so that it vanishes in the low Mach number regime; it is first order at local extrema and second order in smooth regions. Second order in time is obtained with the stiffly accurate, L-stable LSDIRK2 scheme. We have also proved that the scheme preserves moving contacts between stiffened-gas or Noble–Abel stiffened-gas phases, at uniform temperature when thermal or chemical relaxation is active.

The numerical results confirm these properties: second-order convergence of the velocity and the volume fraction on a smooth two-phase vortex at Mach numbers from 2.4 to 0.024, and of the pressure for well-prepared data, face velocities that are discretely divergence-free to $O(M^2)$ down to $M \approx 10^{-6}$ on the Taylor–Green vortex with a time step independent of the Mach number, and agreement with exact solutions and reference results for interfaces, single-phase and two-phase shock tubes, cavitation, evaporation and depressurisation problems, and two-dimensional shock–bubble, Richtmyer–Meshkov, vapour-bubble compression and underwater-explosion problems, at instantaneous and finite relaxation rates.

Future work will address an extension to higher order in space and time, unstructured meshes, and a velocity disequilibrium model, in which the drag relaxation would enter the same local closure.

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